

World Bank Technical Guidance Note

Microsimulation approaches for the assessment of household welfare impacts of climate change and climate action

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January, 2024

Abstract

This note presents the basic building blocks of microsimulation approaches used to assess the welfare impacts of climate change and climate policy. First, we review the intuition behind dynamic microsimulation. We briefly overview microsimulation types based on static and dynamic aging and discuss baseline data challenges and potential external sources for calibrating future demographic and economic pathways. Then, we present the two variants of "Top-Down" simulation approaches: the so-called *behavioral* approach and the pure *reweighting* approach. As their name highlights, "Top-Down" simulation approaches project a given baseline microdata to match exogenous (and sometimes endogenous) macroeconomic aggregates that reproduce the main trends in demographic and economic changes and allow us to explore the redistributive consequences. We also present two versions of the "Shockwaves" approach to microsimulation, a way of breaking free of macroeconomic calibration that explores all possible scenarios that can be generated from the combination of different demographic and economic pathways and that incorporate a missing part in the "Top-Down" architecture: household specific climate damages. We highlight their strengths and challenges along with our presentation of all alternatives. Finally, we summarize our findings and provide recommendations for applications and future research.

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1 Introduction

Climate change is expected to infringe significant challenges on countries' economic, social, and environmental systems. On the one hand, climate change is expected to increase the frequency and intensity of climate-related hazards, causing non-negligible damages to economic and non-economic goods and services, and therefore, affecting not only the aggregate output levels but also the distribution of output and derived welfare among households within a country (ADD REFERENCES). On the other hand, transition policies, the economic shifts aimed at mitigating and adapting to the impacts of climate change, are also very likely to produce a significant redistribution of opportunities and welfare among different economic sectors and different types of workers (ADD REFERENCES). There has been considerable interest among academics and policymakers to estimate the global and regional costs of climate change and policies in terms of GDP losses -(Barrage and Nordhaus (2023)), (Tol (2018)), (Nordhaus (2018)), - and also in terms of welfare measures of poverty and inequality – (Fujimori, Oshiro, Hasegawa, Takakura, and Ueda (2023)), (Hallegatte, Fay, Barbier, and Hochard (2018)), (Hallegatte and Rozenberg (2017)). However, it is unlikely that an aggregate approach will fully capture the impact of climate change on people's well-being. The distribution of climate impacts across households within a given country is as heterogeneous as the distribution of global climate impacts across countries. Using global, regional, or national aggregates will hide the significant difference in effects across diverse areas and communities depending on their exposure and vulnerability to climate hazards. Microsimulations are, arguably, the most appropriate tool to evaluate how climate change and climate policy might affect the distribution of household welfare, taking into account the considerable heterogeneity of households' demographic, geographic, and economic characteristics. Although initially designed for other areas of economic and social research, microsimulations are now often used to assess the distributional impact of climate change and policies within a given country. – ((Van Ruijven, O'Neill, & Chateau, 2015)) and (Hynes and O'Donoghue (2014)), discuss and summarize multiple applications of this technique to country-specific climate challenges. This note describes the structure of three general types of microsimulation models that have been adopted by World Bank staff for the assessment of poverty and inequality consequences of climate change policies at the country level. These methods are mentioned in the guidance document "World Bank Group Macroeconomic Models for Climate Policy Analysis." The three methods -here labeled "reweighting," "behavioral," and "shock waves"- are described within a unified framework to highlight their similarities and differences. This description of the three models intends to guide practitioners about methodological assumptions, required data sources, and suitable implementation conditions according to country characteristics and research program. In our view, the selection of the most appropriate model will depend mainly on the objective of the exercise, the particular data quantity and quality available, and implementation choices that balance the trade-offs between what we need to know, what we know, and what we know we don't know being conscious that projecting into the long-term future potentially has many unknowns that we don't know.

This note is organized into four additional sections. Section 2 provides some background and basic intuition behind the microsimulation of the welfare effects of climate change and climate action, as well as an overview of the three types of microsimulation approaches used in the past. Section 3 presents two varieties of the so-called "Top-down" approach: 1) the reweighting approach and 2) the behavioral approach. "Top-down" microsimulations project household distributions to be consistent with predetermined future pathways of demographic changes and outputs of macroeconomic models in terms of economic growth - and their associated structural changes in the employment and wage structure - and sometimes households' and firms' responses to such changes. Section 4 presents the alternative "shockwaves" approach. The shockwaves approach breaks free from macro-alignment to predefined demographic and economic pathways - and their implications for changes in the employment and wage structures, and simulates an extensive range of possible future scenarios combining demographic pathways with ad-hoc employment and wage structures. Section 5 discusses the advantages and disadvantages of both Top-Down approaches and the intrinsic differences between Top-Down and Shockwaves approaches. Section 6 summarizes the main conclusions and presents the main lessons learned for practitioners and potential areas of improvement that would be worth exploring further.

2 Simulation building blocks

2.1 Background

Microsimulation is an approach to use micro-unit based data to answer "what-if" questions that, otherwise, cannot be answered" [Orcutt (1957)]. Microsimulation consists of using computer programs and microdata (i.e., surveys, censuses, or administrative data from individuals, households, or firms) to assess the potential or counterfactual impact of a given policy upon population welfare or some other socioeconomic variable of interest. As such, microsimulations serve as a "computer experiment" that allows for an assessment of existing or potential policies and ultimately helps better design future economic and social policies. Microsimulation exercises are a standard instrument of policy analysis among academics and policymakers. Several literature surveys review the evolution and extension of these techniques applied to different fields and for various purposes. The literature on using specific microsimulation techniques to evaluate existing and proposed policies is also extensive. World Bank publications surveyed different microsimulation techniques that evaluated distributional impacts - poverty in particular- of economic and social policies. For instance, (Bourguignon and Da Silva (2003)) classifies microsimulation into several types: ex-ante or ex-post, static or dynamic, and partial or general equilibrium. Ex-ante simulations are performed before a program is introduced to predict its likely impact, whereas ex-post simulations assess the impact if the program had been implemented differently. The static (sometimes called accounting) simulations modify the values of variables under study for selected observations in the sample and then assess changes in welfare aggregates (or some other target variable) without any reactions on the part of agents. In contrast, dynamic (sometimes called behavioral) simulations introduce a protocol of reactions to these changes by economic agents. Partial equilibrium simulations constrain the assessment of variables and behaviors of a given market in the short term. In contrast, general equilibrium simulations include reactions in prices and quantities in other markets not directly affected by the policy reform but eventually impacted through medium- and long-term effects across the whole economy. Microsimulation models have grown in extension and complexity in recent years, now including new areas of policy analysis. A more recent classification of microsimulations by (O'Donoghue (2014)) defines four dimensions of complexity that these exercises can include, leading to multiple varieties of models. First, the population complexity refers to the units of analysis included (e.g., individuals, households, firms, farms) and the type of heterogeneity within these groups (e.g., gender, age, size, location) to be considered for policy evaluation. Second, behavioral complexity refers to whether the exercise will include an explicit modeling of behavior by units of analysis. Namely, whether individuals would change labor supply behavior to changes in transfers, consumption patterns to changes in VAT, firms change operations to changes in tariffs, etc. Third, policy complexity, in which realms of policy intervention are to be assessed. Most microsimulation exercises have focused on taxes and transfers, but the discipline is moving toward assessments of transportation, health, education, and environmental policies. Fourth, these models can also venture into time and space when modeling changes over time or location. Models about demographic change, fertility, pensions, and educational policies need an explicit definition of dynamics over the long term. In contrast, models on workers commuting and transportation/firm location need geographical data. Another way to classify microsimulation approaches -introduced initially by ...- are by 1) static aging and 2) dynamic aging. Static aging refers to the process by which individuals or households in a survey change their representativeness of a population over time, keeping their characteristics, attributes, or behaviors unchanged. For example, static aging would mimic a process of growth in employment and a structural change (that moves individuals out of non-employment and agriculture into industry and services) by only changing the relative importance of those individuals (or their households), keeping all other observed characteristics (i.e., labor earnings or sector of employment) fixed or unchanged for the duration of the simulation. In other words, static aging is adjusting households' importance (weights) to align simulated survey aggregates in baseline to exogenous (or endogenous) macro forecasts. Notice that in static aging, there is no modeling of the process that drives these changes. Two important points are worth noticing. First, the static aging approach does not change any household characteristics. It only changes its relative importance in the sample by increasing or decreasing their weights. Second, the change in weight occurs not at the individual level but at the household level. Therefore, differences in the relative importance of a given household in the overall distribution of the variable of interest (e.g., schooling) will also alter the overall distribution of all other variables. In turn, dynamic aging refers to the process that changes the characteristics of individuals in a

population over time by applying a modeling rule (exogenous or endogenous). In simple words, dynamic aging aligns baseline data with exogenous projections by changing the attributes of individuals (or households) using a simple or complex rule -instead of just changing their weights. Using the same example of aligning our baseline survey schooling distribution with a target distribution in 2050 that accounts for improvements in education will change the observed characteristic of individuals (not households) based on ad-hoc (arbitrary) or behavioral (estimated) rules that condition and identify who is more likely to gain more schooling. This is useful when the focus is on understanding the current state of the population without considering how it may evolve in the future. As we will analyze in detail in the next section, the choice between static and dynamic aging will depend not only on the research or policy questions being addressed, the level of detail needed in the simulation, and, in particular, on the quantity and the quality of the available data for the baseline. The three models described below can be cataloged according to these different categories. They are ex-ante microsimulation models because they attempt to model the future distribution of welfare impacts due to climate change. In all cases, the unit of analysis will be individuals and households (not firms or farms). Policy intervention will only refer to climate change and climate policy interventions. It will be shown that some of them take a static stance, whereas others take a behavioral approach, with or without general equilibrium underpinnings. The static/dynamic aging approach to the different components of the model will be a critical element of the unified approach to describe these three models in what follows.

2.2 Household income-generation and baseline data

Formally, counterfactual microsimulations are usually interested in the household's welfare differential, $d_{h,s}(\cdot)$, when we compare a "what-if" counterfactual scenario, $y_{h,s}(\cdot)$, with an observed baseline scenario, $y_{h,b}(\cdot)$,

$$d_{h,s}(\cdot) = y_{h,s}(\cdot) - y_{h,b}(\cdot) \quad (1)$$

where $y_{h,b}$ stands for a welfare measure, for household "h", under a counterfactual scenario "s" or a baseline "b".

Microsimulations may also be interested in a household's welfare change over time $d_{h,t}$, when we compare the income distribution that would prevail in the near future, $y_{h,t}$, with an observed baseline year in the recent past, $y_{h,0}(\cdot)$,

$$d_{h,t}(t) = y_{h,t}(\cdot) - y_{h,0}(\cdot) \quad (2)$$

Both analyses are static microsimulation models because they do not account for long-term trends in the population structure, such as age-sex composition, human capital accumulation, internal migration, or even labor supply/consumption expenditure reactions to policy changes. Even when they do, changes in the population are usually not large enough to produce significant changes in the observed baseline composition, at least when compared to those that would be expected in the long run.

On the other hand, long-term microsimulation models need to radically change the observed baseline society to represent (or match) exogenous or endogenous structures that are expected in the future. Microsimulations applied to assessing household welfare consequences of climate change and climate policy are necessarily long-term. Climate change impacts are not only more likely to occur but are expected to be more significant in the long-term future. Therefore, we are interested in comparing household's welfare differentials $d_{h,st}(\cdot)$, when we compare a counterfactual income distribution that would prevail in a long-run future "t" under a given climate scenario, "s," $y_{h,st}$, with that one that would prevail in the long-run future under a baseline scenario, $y_{h,b0}$.

$$d_{h,st}(\cdot) = y_{h,st}(\cdot) - y_{h,b0}(\cdot) \quad (3)$$

Notice that welfare comparisons due to climate change are made in the long-run period "T," usually 20 to 30 periods from now. A period so far from now - or from the earliest observed baseline data available, creates the need to use dynamic microsimulation to update the current structure of the society as is observed in the baseline, with the forecasted structure that is expected in the future.

The difference in welfare measures over time and across scenarios requires a definition of the welfare aggregate. Often, a welfare aggregate is defined as a household income-generating identity. Because the labor market is usually the most significant source of income for most households, particularly in developing countries, the income-generating identity usually distinguishes at least two sources of individual income. Let's

$w_{ih,st}$ and $z_{ih,st}$ be the labor earnings and other non-labor sources of income non-labor earnings that may include capital income (from financial or physical assets), household transfers (including remittances), and government transfers of individual i in household h , under scenario s at time t ,

$$y_{ih,st} = w_{ih,st} + z_{ih,st} \quad \text{for } i \in h \quad (4)$$

Given a survey with information about these income sources for an individual in a given household, then the distribution of three variables can fully describe total household income at the baseline for the total population under a given scenario s , and at the time, t ,

1. the household-specific frequency sampling -shared by all household members, $\theta_{ih,st}$;
2. individual labor earnings, $w_{ih,st}$; and
3. individual (or household) non-labor income $z_{ih,st}$, which may also include the net balance of taxes and transfers (hence equation 4 would refer to disposable income).

Total household income is only the aggregation of both sources of income across all household members, index by i , across all households, index by h .

$$\begin{aligned} Y_{st} &= \sum_h \sum_{i \in h} y_{ih,st} \cdot [\theta_{ih,st}] \\ &= \sum_h \sum_{i \in h} [w_{ih,st} + z_{ih,st}] \cdot [\theta_{ih,st}] \end{aligned} \quad (5)$$

Therefore, a microsimulation of a counterfactual household income distribution will be given by estimates of new labor earnings and non-labor incomes under a given scenario s at a target period, t , and, assuming a *closed model*, a new set of household weights, $\theta_{ih,st}^*$ ¹.

A more detailed formulation of the household income-generation identity that accounts for differences between the type of workers, indexed by q , and sectors of activities, indexed by r , could be formulated as:

$$\begin{aligned} Y_{st} &= \sum_h \sum_{i \in h} y_{ih,st} \cdot [\theta_{h,st}] \\ &= \sum_h \sum_{i \in h} \left[\sum_q \sum_r l_{ih,st}^{qr} \cdot w_{ih,st}^{qr} + z_{ih,st} \right] \cdot [\theta_{ih,st}^*] \end{aligned} \quad (6)$$

where $l_{ih,st}^{qr}$ is an indicator variable that takes value 1 when individual i , of household h , in scenario s and period t , is a worker type q in sector r , and earns $w_{ih,st}^{qr}$. Workers' type may include a disaggregation by sex and skill. Employment sectors include non-employment and a convenient disaggregation of economic sectors (e.g., Agriculture, Industrial, and Services) and -sometimes, employment types (public and private sector wage salaried workers or self-employed).

Rather than simply aggregating total incomes, this formulation of individual-household incomes may also serve to define a social welfare function defined as a vector of individual-household welfare that is a function of income as a welfare aggregate:

$$U_{s,t} = U(u_1(y_{1,1,s,t}), u_2(y_{2,1,s,t}), \dots, u_I(y_{I,H,s,t})) \quad (7)$$

A cardinal formulation of the social welfare function $U(\cdot)$ may focus on, or be sensitive to, inequality or poverty or some other formulation of social welfare. Then microsimulation may render counterfactual estimates of changes in poverty, inequality, or some other welfare measure across scenarios and overtime -beyond mere changes in incomes as described in equation 3- as follows:

¹ *Closed* microsimulation models do not allow for existing individuals (or households) to exit the sample or for new individuals (or households) to enter the sample. *Open* microsimulations - models that allow individuals (and households) to exit or enter the sample via resampling techniques. This latter type is much less common than the former.

$$D_{s,t} = U_{s,t}(\cdot) - U_{b,t}(\cdot) \quad \text{for } t \in \{1, 2, \dots, T\} \quad (8)$$

If the formulation is, in addition, decomposable by population groups, then welfare changes by groups can be defined as follows:

$$D_{g,st} = U_{g,st}(\cdot) - U_{g,bt}(\cdot) \quad \text{for } t \in \{1, 2, \dots, T\}, g \in \{1, 2, \dots, G\} \quad (9)$$

where $U_{g,st}(\cdot)$ represents the social welfare measure as per equation 7 for a specific group of individuals or households g (e.g., ethnic groups, family types, regional population). This welfare focus and socio-demographic granularity of microsimulation exercises are some of the main contributions of this technique to the economic analysis of policy changes, usually not possible through other, more aggregate methods. Often, microsimulations concentrate on the welfare/poverty/inequality impact of policy changes for the whole society and/or for some specific groups of society.

2.3 Microsimulation as projected changes by components of income

Given the definition of income from equation 6, microsimulation can be seen as a projection of information for a baseline scenario, $s = 0$, on a baseline year, $t = 0$, into counterfactual scenarios and future periods. More specifically, the counterfactual future income distribution for a hypothetical scenario, s , and future year, t , can be described by a n -tuple,

$$[\Psi_{ih,st}[a_{ih,st}, b_{ih,st}, c_{ih,st}, u_{ih,st}]; l_{ih,st}, w_{ih,st}, z_{ih,st}; \theta_{ih,st}] \quad (10)$$

where $\Psi_{ih,st}[a_{ih,st}, b_{ih,st}, c_{ih,st}, u_{ih,st}]$ is a vector that groups demographic characteristics including at least, age $a_{ih,st}$, sex, $b_{ih,st}$, schooling, $c_{ih,st}$, and place of residence, $u_{ih,st}$; $l_{ih,st}$ employment status and occupational choice; $w_{ih,st}$ labor earnings; $z_{ih,st}$ non-labor earnings and net taxes; $\theta_{ih,st}^*$ importance weights that guarantee the alignment of the microsimulation output with forecasted macroeconomic aggregates.

Microsimulation involves at least three steps

- Population alignment by particular population partitions,
- Employment alignment by industry (and worker type), and
- Wage alignment by industry (and worker type),

Occasionally, an additional step of per-capita household income growth alignment is added to guarantee consistency between the simulated and forecasted monetary welfare aggregates.

The first step, aligning microsimulations with population aggregates and marginal or joint distributions, usually uses static aging. It changes the value of baseline importance weights, $\theta_{ih,st}^*$, to align microsimulations with specific counterfactual demographic pathways that characterize the evolution of the demographic structure. These counterfactual pathways are usually exogenously given. Annex A, at the end, presents the commonly used "Shared Socioeconomic Pathways" (SSP) database, which provides the usual source of population projections by age, sex, schooling level, and place of residence for all countries up to 2100².

Once the counterfactual demographic pathway is chosen, the second and - arguably, the most important part of the microsimulation is the alignment with a counterfactual employment structure in terms of employment status, i.e., whether individuals are economically active, employed, or unemployed, and also in terms of their particular sector or occupation "choice." Different climate scenarios, due to damage or transition to net zero, will generate employment reallocation in the future. On the one hand, the static aging approach assumes reallocation occurs at random and simply follows the same procedure of adjusting the household importance weights, $\theta_{ih,st}$, keeping the employment and wage values observed at the baseline year, $(l_{ih,b0}, w_{ih,b0})$ labor earnings; . On the other hand, the dynamic aging approach assumes it is possible to estimate (not the mean but) the full range of behavioral responses so that the new counterfactual distribution

²Even at the baseline scenario period, $s = b, t = 0$, there might exist discrepancies between the population structure obtained from the household survey, $\{(a_{c,b0} \cdot \hat{b}_{c,b0} \cdot \hat{e}_{c,b0}), \hat{u}_{b0}\}$, and the population structure observed from external sources $\{(\hat{a}_{c,b0} \cdot \hat{b}_{c,b0} \cdot \hat{e}_{c,b0}), \hat{u}_{b0}\}$.

of employment and wages can be predicted, at least for those labor market status or industry, $(l'_{ih,st}, w'_{ih,st})$ - where the apostrophe, ', is used to emphasize their counterfactual nature.

The third step is aligning the simulated wage structure with counterfactual future projections. A critical difference between the "Top-Down" approaches to microsimulation and the Shockwaves approach is whether this stage will align or not with forecasted macroeconomic (average or total) wage bill growth. Both top-down microsimulations, the reweighting and the behavioral, calculate residual growth rates that adjust for the additional growth necessary to match the forecasted labor productivity growth. Demographic changes such as population aging, urbanization and, more fundamentally, accumulation of schooling, and structural change reallocation are expected to increase average productivity so that to guarantee a perfect alignment, only a residual rescaling factor, $\hat{\delta}_{Wst}$, is necessary to align microsimulations with the average labor productivity growth $\hat{g}_{Wst}$. On the other hand, the shockwaves approach compounds estimates of the average labor productivity growth $\hat{g}_{Wst}$, with those already observed due to the changes in the demographic composition and labor reallocation.

Table 1 attempts to summarize the main differences between the three approaches described in this paper regarding the two main mechanisms used to forecast baseline data and align it with forecasted aggregates. All three models used static aging - changing the importance weights of households to account for changes in the demographic structure in the future. As we explain in the following sections, there might be specific implementation choices regarding the particular reweighting algorithms used, the number of population partitions, or where the reweight is done to match marginal or joint distributions, but the mechanism is the same. The reweighting and shockwaves approaches also use static aging to account for changes in the employment structure by labor status and sector. Changing weights at the household level to account for joint changes in the demographic and employment structure not only assumes a random reallocation process but can be challenging to implement if the sample size of the baseline survey is small or there are particular groups underrepresented in it. On the other hand, the behavioral model uses dynamic aging to account for both employment reallocation and counterfactual wages for those who move into employment or change sectors. This approach not only requires good-quality data but also implicitly assumes that some identification assumptions are met to conduct a piecewise estimation. Finally, both Top-down approaches, the reweighting and behavioral approaches, use residual growth rates to align simulated growth in wages with forecasted growth - adjusting only for the additional growth necessary to match the forecasted labor productivity growth - column(4). On the other hand, the shockwaves compound the labor productivity growth with those already observed due to changes in the demographic and employment composition of the simulation. All three models use forecasted aggregate growth to update household non-labor earnings. Although not included in Table 1, in some applications, it might be helpful to align labor further so that the growth in total household labor earnings also coincides with forecasted aggregated growth from the macro model.

Table 1: A summary of microsimulation assumptions and mechanisms

Approach	Mechanisms				
	Demographic	Aging Employment	Labor earnings	Labor earnings	Non-labor earnings
Reweighting	$(\psi_{ih,b0} \theta_{h,st}^*)$	$(l_{ih,b0} \theta_{h,st}^*)$	$(w_{ih,b0} \theta_{h,st}^*)$	$(\delta_{st} \theta_{h,st}^*, \hat{g}_{Wst})$	$\hat{g}_{Yst}$
Behavioral	$(\psi_{ih,b0} \lambda_{h,st}^*)$	$(l'_{ih,st} \lambda_{h,st}^*)$	$(w'_{ih,st} \lambda_{h,st}^*)$	$(\delta_{st} \lambda_{h,st}^*, \hat{g}_{Wst})$	$\hat{g}_{Yst}$
Shockwaves	$(\psi_{ih,b0} \tilde{\theta}_{h,st}^*)$	$(l_{ih,b0} \tilde{\theta}_{h,st}^*)$	$(w_{ih,b0} \tilde{\theta}_{h,st}^*)$	$\tilde{g}_{Wst}$	$\tilde{g}_{Yst}$

Another way to make differences between simulation approaches is to write down the simulated total household income for a scenario, s , at a time, t , as a function of both importance weights and rescaling parameters. New weights can adjust for both demographic and employment changes, $\theta_{h,st}^*$, for only demographic changes, $\lambda_{h,st}^*$, or for a random draw that combines alternative scenarios of demographic and employment changes, $\tilde{\theta}_{h,st}^*$. Rescaling parameters can adjust for either residual growth in labor productivity, δ_{Wst}^{qr} - the remaining growth necessary after accounting for observed labor productivity growth due to changes in the demographic and employment composition of employment, or overall labor productivity growth, g_{Wst} ,

that adds up to the growth due to employment composition.

In the reweighting approach - equation 5.1, total simulated household income for each scenario, s , and at target year t , Y_{st}^R , depends on new weights that align demographic and employment variables, $\theta_{.h,st}^*$, and a *residual* rescaling factor $\delta_{W_{st}}^{qr}$ that aligns to forecasted labor productivity growth by sector and worker type.

$$\begin{aligned}
Y_{st}^R &= \sum_h \sum_{i \in h} y_{ih,st} \cdot [\theta_{.h,st}^*] \\
&= \sum_h \sum_{i \in h} [l_{ih,st} \cdot w_{ih,st} + z_{ih,st}] \cdot [\theta_{ih,st}^*] \\
&= \sum_h \sum_{i \in h} \left\{ \left[\sum_{q=1}^Q \sum_{r=1}^R l_{ih,b0}^{qr} \cdot w_{ih,b0}^{qr} \cdot \delta_{st}^{qr} \right] + [\hat{g}_{Y_{st}} \cdot z_{ih,b0}] \right\} \cdot [\theta_{.h,st}^*]
\end{aligned} \tag{11}$$

In the behavioral approach - equation 5.1, total simulated household income for each scenario, s , and time period t , Y_{st}^B , depends on weights that align only demographic changes, $\lambda_{.h,st}^*$, and new *counterfactual* distributions of employment and wages ($l'_{ih,b0}, w'_{ih,st}$), and a *residual* rescaling factor, $\delta_{W_{st}}^{qr}$.

$$\begin{aligned}
Y_{st}^B &= \sum_h \sum_{i \in h} y_{ih,st} \cdot [\lambda_{.h,st}^*] \\
&= \sum_h \sum_{i \in h} [l'_{ih,b0} \cdot w'_{ih,st} + z_{ih,st}] \cdot [\lambda_{ih,st}^*] \\
&= \sum_h \sum_{i \in h} \left\{ \left[\sum_{q=1}^Q \sum_{r=1}^R l'_{ih,b0}{}^{qr} \cdot w'_{ih,b0}{}^{qr} \cdot \delta_{st}^{qr} \right] + [\hat{g}_{Y_{st}} \cdot z_{ih,b0}] \right\} \cdot [\lambda_{.h,st}^*]
\end{aligned} \tag{12}$$

Finally, in the Shockwaves approach - equation 13, total simulated household income for each scenario, s , at target year t , Y_{st}^S , depends on new weights that align the baseline with a particular *draw* of alternative demographic and employment scenarios, $\tilde{\theta}_{.h,st}^*$, and a particular drawn of labor productivity overall growth for each economic sector and worker type, $\tilde{g}_{W_{st}}^{qr}$ that **do not** align to a particular forecast of macroeconomic aggregates. In Section 4, we provide details on how the shockwaves randomly select counterfactual parameters among a finite but significant set of alternative scenarios to construct not only one particular future but a large set of alternative futures.

$$\begin{aligned}
\tilde{Y}_{st}^S &= \sum_h \sum_{i \in h} y_{ih,st} \cdot [\tilde{\theta}_{.h,st}^*] \\
&= \sum_h \sum_{i \in h} [l_{ih,st} \cdot w_{ih,st} + z_{ih,st}] \cdot [\tilde{\theta}_{ih,st}^*] \\
&= \sum_h \sum_{i \in h} \left\{ \left[\sum_{q=1}^Q \sum_{r=1}^R l_{ih,b0}^{qr} \cdot w_{ih,b0}^{qr} \cdot \tilde{g}_{W_{st}}^{qr} \right] + [\tilde{g}_{Y_{st}} \cdot z_{ih,b0}] \right\} \cdot [\tilde{\theta}_{.h,st}^*]
\end{aligned} \tag{13}$$

3 The "Top-Down" approach

3.1 The reweighting approach

The *reweighting* approach has been adopted to assess the welfare impact of climate change and climate policy in several countries with different economic structures and exposures to climate change. An early adoption was by [(Buddelmeyer et al., 2012)], which is usually cited in subsequent studies. They assess the changes in net incomes and Gini coefficients by quintile and family type in Australia between 2010 and 2030 for a baseline and two alternative decarbonization scenarios. Later, [(Vandyck & Van Regemorter, 2014)] use a CGE model and a version of EUROMOD to assess the impact of carbon taxation upon consumer prices and

several sources of household incomes, with different compensating mechanisms, such as lump-sum transfers or reduced labor taxes, across geographic regions in Belgium. More recently, [(Montaud et al., 2017)] uses a CGE model about the impact of climate change on crop yields, as well as the coping strategies associated with investment in irrigation capacity and modern crop varieties, upon poverty and food security indicators in Nigeria over 25 years. The welfare impacts are calculated with a microsimulation model linked to the CGE model by a reweighting procedure like the one from [(Buddelmeyer et al., 2012)].

The *reweighting* approach refers to a microsimulation that uses static aging - changes in household importance weights to account for changes not only in the main demographic components (population numbers, composition by sex and age, schooling distribution, and urbanization) but also to account for changes in the employment. A *reweighting* microsimulation usually has five steps: 1) aligning baseline microdata with future demographic projections and macroeconomic forecasts of the employment structure by changing household weights, 2) rescaling baseline wage distributions to align them with macroeconomic forecasts of the aggregate wage bill by economic sector (and, sometimes, by workers' type), 3) rescaling non-labor usually assuming a rate of growth equal to macroeconomic forecasts of GDP or consumption growth. Sometimes, a further rescaling of total household income is made to align growth rates of simulated total household income with macroeconomic forecasts of the overall growth rate.

3.1.1 Aligning with future demographic and employment structure

To align the baseline survey population structure with a projected demographic structure and a forecasted employment structure the reweighting approach uses a static aging approach. Static aging calculates a new set of household importance weights that allow us to align baseline data with future demographic and employment structures. Formally, let H_{b0} be the total population at the baseline scenario and year, and $\theta_{ih,b0}$ be the initial frequency weight of individual i within household h so that, $H_{b0} = \sum_h \sum_i \theta_{ih,b0}$.

Total population at the baseline scenario and year can be written as ³. Since in most household surveys households are the final unit of sampling, sampling weights attached to every individual needs to be equal within the household ⁴.

$$\theta_{ih,b0} = \theta_{jh,b0} = \theta_{.h,b0} \quad \text{for } i, j \in h : i \neq j$$

Let the projected demographic structure of the population be given by a n -tuple that describes population partitions of age $\hat{a}_{st}$, sex, $\hat{b}_{st}$ schooling, $\hat{c}_{st}$ and place of residence, and let $\hat{e}_{st}$ describe the employment status, and sector population partition, $\hat{e}_{st}$. Formally,

$$[\hat{a}_{st} \cdot \hat{b}_{st} \cdot \hat{c}_{st}; \hat{u}_{st}; \hat{e}_{st}]$$

so that

$$\hat{H}_{st} \cdot 1 [\hat{a}_{st} \cdot \hat{b}_{st} \cdot \hat{c}_{st}; \hat{u}_{st}; \hat{e}_{st}] = \sum_h \sum_i \tilde{\theta}_{ih,st} 1 [\hat{a}_{st} \cdot \hat{b}_{st} \cdot \hat{c}_{st}; \hat{u}_{st}; \hat{e}_{st}] \quad (14)$$

where $1[\cdot]$ is the indicator function, $\tilde{H}_{st}$ is the new total population, and each member i of household h has the same weight,

$$\tilde{\theta}_{ih,st} = \tilde{\theta}_{jh,st} = \tilde{\theta}_{h,st} \quad \text{for } i, j \in h : i \neq j$$

³Or, in terms of probabilities, as

$$1 = \sum_h \frac{\theta_{h,b0}^*}{H_{b0}} = \sum_h \vartheta_{h,b0}^*$$

,

$$H_{b0} = \sum_h \theta_{ih,b0} \cdot \text{household size}_h = \sum_h \theta_{h,b0}^*$$

⁴For a discussion of survey design and population weights, for the analysis of household surveys and development policy see [(Deaton, 2019)]

Total population at counterfactual scenario s and the target year t can be written as ⁵,

$$\sum_h \tilde{\theta}_{.h,st} \text{ household size}_h = \sum_h \tilde{\theta}_{h,st}^* = \tilde{H}_{st}$$

To find a distribution of household weights, $\theta_{.h,st}^*$ (or $\vartheta_{.h,st}^*$ in probability terms) from the initial distribution of household weights ($\theta_{.h,b0}^*$) (or $\vartheta_{.h,b0}^*$ in probability terms) we impose a set of constraints over different population partitions - in this particular scenario under three populations partitions defined by: joint distributions of gender, age, and schooling, marginal distributions of urbanization, and marginal distributions of employment status and sector choice, that will define the growth rates of importance weights for each population partition. The new distribution of household weights can be obtained by solving the following optimization problem,

$$\min \Gamma(\vartheta, \tilde{\vartheta}) = \sum_h \tilde{\vartheta}_{.h,st}^* \ln \left(\frac{\tilde{\vartheta}_{.h,st}^*}{\vartheta_{.h,b0}^*} \right) \quad (16)$$

subject to,

$$\begin{aligned} E(X_{kst}) &= \sum_h \vartheta_{.h,b0}^* \bar{x}_{kst} & k &= 1, \dots, K \\ \sum_h \vartheta_{.h,b0}^* &= 1 \end{aligned} \quad (17)$$

where $\bar{x}_{kst}$ is the mean of the x_k characteristic within household h . This procedure guarantees that the distribution $\tilde{\vartheta}_{.h,st}^*$ satisfies all moment constraints and the normalization restriction at the same times that guarantees that we deviate as little as possible from the initial distribution ($\vartheta_{.h,b0}^*$).

The solution to this problem is given by,

$$\begin{aligned} \hat{\vartheta}_{.h,st} &= \vartheta_{.h,b0}^* \frac{\exp(X\hat{\mu})}{\Omega(\hat{\mu})} \rightarrow \sum_h \hat{\vartheta}_{.h,st} = \sum_h \vartheta_{.h,b0}^* \varrho_h = \tilde{H}_{st} \\ \text{with } \varrho_h &= \frac{\exp(X\hat{\mu})}{\Omega(\hat{\mu})} \end{aligned} \quad (18)$$

where $\hat{\mu}$ is the Lagrange multiplier for the constraint $E(X)$, and Ω is the normalizing factor to scale the new population to the target size.

3.1.2 Alignment with changes in the wage structure (by sector and workers' type)

With a new set of importance weights that aligns the simulation with the counterfactual future demographic and employment structure, the next step is to adjust the wages - ideally by workers' type, indexed by q (e.g., skill level) and industry, indexed by r . To align baseline labor earnings with the counterfactual forecasts of the macroeconomic model, we need to calculate a *residual* growth in labor productivity, δW_{st}^{qr} , that guarantee that, for each workers' type, q , and each industry, r , the simulated average wage growth, g_{st}^{qr} , aligns with the forecasted macro growth between counterfactual scenario s at target year t and the baseline scenario b at the initial period 0, $\hat{g}_{st}^{qr}$.

$$\hat{g}_{st}^{qr} = \hat{w}_{st}^{qr} / \hat{w}_{b0}^{qr} \quad (19)$$

where $\hat{w}_{st}^{qr}$ and $\hat{w}_{b0}^{qr}$ represent the forecasted values coming from the macro model for the counterfactual scenario s at target year t , and the baseline scenario b and the initial year 0, respectively.

It is essential to mention that because of demographic realignment and employment realignment, very likely the simulated distribution of labor earnings in scenario s at target year t will already be different from its

⁵In probability terms,

$$1 - \sum_h \frac{\tilde{\theta}_{.h,st}^*}{\tilde{H}_{st}} = \sum_h \tilde{\vartheta}_{.h,st}^* \quad (15)$$

baseline counterpart. For example, accounting for population aging, increasing human capital accumulation, and urbanization is very likely linked to an increase in the share (the importance weights) of those with relatively higher labor earnings - and their associated decrease in the share of those with relatively lower labor earnings.

Let $\bar{w}_{st}^{qr}$ and $\bar{w}_{b0}^{qr}$ be the average labor earnings for counterfactual scenario s and period t , and the average labor earnings at the baseline scenario b and base year 0. It is easy to see that the simulated average labor earnings by workers' type and sector only after reweighting will differ from baseline average weights because the differences in composition due to demographic and employment structure changes accounted for the new importance weights. In particular,

$$\bar{w}_{b0}^{qr} = \frac{\sum_i \sum_{i \in h} (1_{ih,b0}^{qr} \cdot w_{ih,b0}^{qr} \cdot \theta_{ih,b0}^*)}{\sum_h \sum_{i \in h} (1_{ih,b0}^{qr} \cdot \theta_{ih,b0}^*)} \quad (20)$$

$$\bar{w}_{st}^{qr} = \frac{\sum_h \sum_{i \in h} (l_{ih,b0}^{qr} \cdot w_{ih,b0}^{qr} \cdot \theta_{h,st}^*)}{\sum_h \sum_{i \in h} (l_{h,st}^{qr} \cdot \theta_{ih,st}^*)} \quad (21)$$

In general, the new weights will not be enough to make the wage ratios $\bar{w}_{st}^{qr}/\bar{w}_{b0}^{qr}$ align to their macro forecast, $g_{W_{st}}^{qr}$. So, there is a need for residual growth in average labor productivity that aligns microsimulations with macro forecasts. Therefore, the residual growth in labor productivity will be implicitly defined by,

$$\hat{g}_{W_{st}}^{qr} = \delta_{W_{st}}^{qr} \cdot \bar{w}_{st}^{qr}/\bar{w}_{b0}^{qr} \quad (22)$$

To the risk of sound repetitively, notice that the simulations do not adjust baseline wages using macro forecast of the overall labor productivity growth, $\bar{w}_{st}^{qr}$, but only a residual productivity growth, $\delta_{W_{st}}^{qr}$, that accounts for the part of labor productivity growth not accounted by labor productivity growth already accounted for changes in the demographic and employment composition of the labor force. Also notice that, although macro-to-micro alignment under this approach guarantees a perfect alignment between simulated and forecasted growth rates across industries and workers' types, it does not guarantee the alignment of levels ⁶

3.1.3 Alignment to overall economic growth

After rescaling to account for forecasted labor productivity growth - by sector and workers' type, there may be a need to adjust simulation counterfactuals further to be consistent with forecasted aggregate growth of total (or per capita) income or consumption. To guarantee micro-macro consistency even when the simulations may ignore or underestimate some forms of income, such as capital income, all simulated sources of household incomes are adjusted using the same rescaling factor in a distribution-neutral way. This assumption is reasonably accurate, particularly for poor households, given the small capital income they obtain. For wealthier households, the margin of error might be more significant. However, even for these households, which tend to be more skilled rather than unskilled labor, earnings tend to play a more critical role [Bussolo, 2013].

To align simulated growth rates in per-capita income with the macro forecasts, an **overall income rescaling** factor, $\gamma_{Y_{st}}$, that ensures that the simulated growth rate of the average counterfactual household income in the survey is aligned to their macro forecast counterpart in all years can be defined as follows.

$$\hat{g}_{Y_{st}} = \gamma_{Y_{st}} \cdot \bar{y}_{st}/\bar{y}_{b0} \quad (23)$$

where $\hat{g}_{Y_{st}} = \bar{y}_{b0}/\bar{y}_{st}$ is the macro forecast of per capita income (or consumption) growth between scenario s at time t and the baseline scenario b at initial year 0, and $\bar{y}_{b0}$ and $\bar{y}_{st}$ are, respectively, baseline per capita income in the baseline scenario b at initial year 0, and the simulated percapita income in counterfactual scenario s at the target year t ⁷.

⁶An alternative approach would be to guarantee the consistency between simulated and forecasted levels at the expense of the alignment of growth rates.

⁷Where the counterfactual average per capita income at the baseline scenario b and initial year 0 is given by,

$$\bar{y}_{b0} = \frac{\sum_h \sum_{i \in h} y_{ih,b0} \cdot \theta_{ih,b0}^*}{\sum_h \sum_{i \in h} \theta_{ih,b0}^*} \quad (24)$$

From a substantive point of view, the reweighting model does not hypothesize how household choices among work and leisure react to new output and factor prices or transfers due to social protection policies. It mainly changes household importance weights to account for changes in the relative size of different population groups so that the size and distribution of these groups match the demographic projections and macroeconomic forecast. Consequently, the reweighting model does not allow the study of individual transitions between employment states or sectors. For example, individuals working in agriculture in the baseline year survey will continue working under every scenario and every period. If we are interested in specific population groups affected by certain policies and -more importantly- how they react, the reweighting approach assumes random reallocation among group members and might give a partial answer. Yet, the reweighting approach's main advantage is that it is straightforward to explain. We now turn to some of its implementation limitations.

3.1.4 Implementation choices

Optimal population partitions

From an implementation perspective, the reweighting model may also encounter some other problems. Reweighting cannot be used when there are no sample observations in a particular population partition of interest, or when its sample is underrepresented, so that a very high weight may have to be applied to those observations. Moreover, if the baseline sample is small and there is a need to align many variables and categories, age $\times$ sex $\times$ schooling, reweighting can run into stability/convergence problems or require significant time to complete. Notice that, the restrictions defined by equation 17 that feed into optimization problem are defined as average values at the household level. Balancing household with restrictions pushing importance weights in different directions can also create convergence problems even with large sample sizes. If the composition of a particular household has people in both, the shrinking (e.g. agriculture) and expanding sector (e.g., industry or services), we can not increase the weight of the person in the expanding sector and reduce the weight of their family member in the shrinking sector, because all household members must have the same importance weight by definition. In some cases, the new weights may differ from the original surveys or be difficult to accept from other considerations (e.g., some are too close to zero while others are too large). For a discussion of this problem and potential solutions, see [(Pacífico, 2014)], [(Kolenikov, 2014)] and [(Wittenberg, 2010)].

Reweight over joint or marginal distributions

Reweighting defines growth rates for importance weights for each population partition that satisfies all moment constraints and the normalization restriction at the same times that guarantees that the new weights deviate as little as possible from the initial distribution. Usually, reweighting match moment constraints defined by the joint distributions of gender, age, and schooling, and marginal distributions of urbanization, and employment status and sector choice. That is,

$$[\text{age} \times \text{sex} \times \text{schooling}; \text{urbanization}; \text{employment}]$$

An alternative to this procedure would match moment constraints defined only by the marginal distributions of all variables, i.e.,

$$[\text{age}; \text{sex}; \text{schooling}; \text{urbanization}; \text{employment}]$$

The choice among these alternatives is not trivial. Depending on the number of population partitions, reweighting on the joint distribution may produce a different solution than reweighting only on the marginal distribution of the alignment variables. Figure 1 presents alternative population partitions for a reweighting on only two alignment variables: age and schooling. Notice that the number of restrictions derived from combining these two variables depends on the number of partitions we define in age and schooling. At one extreme, we can explain 21 5-year cohorts of age and three skills levels - defined as progressions of schooling

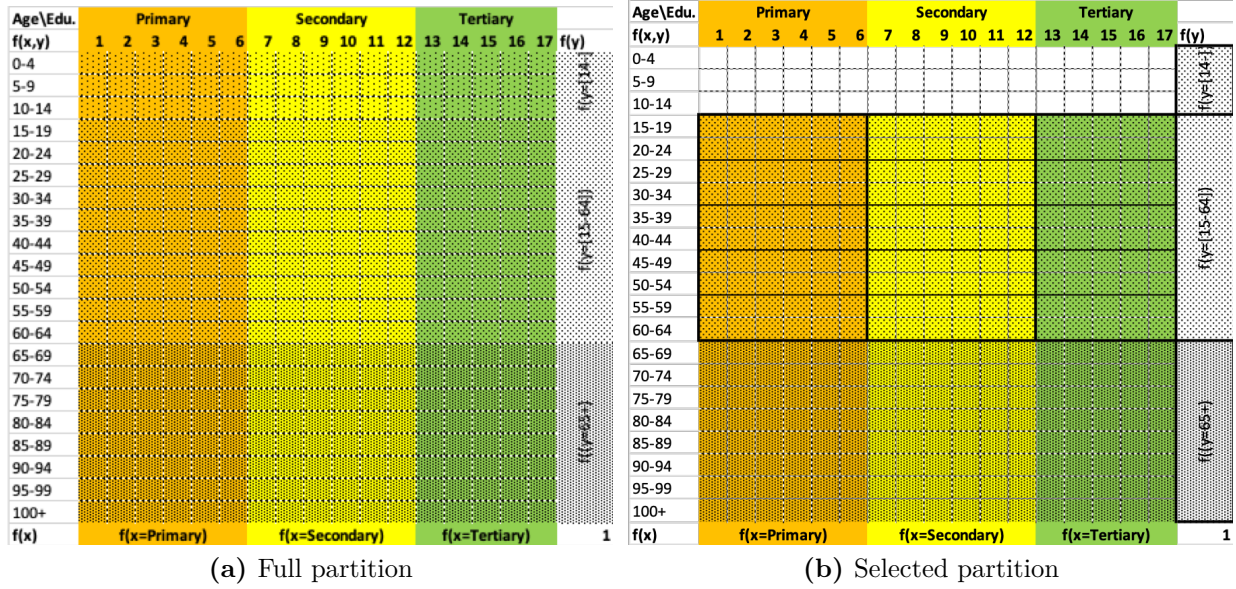
and the counterfactual average per capita income at the counterfactual scenario s at the target year t is given by,

$$\bar{y}_{st} = \frac{\sum_h \sum_{i \in h} y_{ih,st} \cdot \theta_{h,st}^*}{\sum_h \sum_{i \in h} \theta_{h,st}^*} \quad (25)$$

where $y_{ih,st} = w_{st} + \hat{g}_{Yst} \cdot z_{ih,b0}$.

levels, with $(18 \times 3 + 3) = 57$ moment restrictions if we choose to match joint distributions for all persons above 15 years old, and only $(21 + 3) = 24$ moment restrictions if we choose to use the marginals (Panel (a)). Panel (b) illustrates an alternative -a more feasible scenario where age cohorts are defined by 10-year increments and the joint distribution is used only for those of working age, i.e., 15 to 64 years old, with $(5 \times 3 + 2) = 17$ moment restrictions (Panel (b)). The choice of the best population partition will depend on the country's conditions and will determine the quality of the solution. Table 2 illustrates the choice in the number of restrictions with different population partition choices.

Figure 1: Alternative population partitions



Note: $f(x)$ denotes age marginal distribution. $f(y)$ denotes education levels marginal distribution. $f(x, y)$ denotes age, education levels joint distribution.

Reweighting algorithm

Different reweighting algorithms exist⁸. To date, most climate microsimulations applications have followed the GIDD approach that relies on the minimum-cross entropy method proposed by Wittenberg (2010) and optimized with different solvers by Corral and Salcedo (2022) `wentropy` Stata command.

3.2 The "Behavioral" approach

The behavioral top-down microsimulation model is credited to have been proposed initially with applications to distributional impacts of macro and trade policy in (Bourguignon & Pereira da Silva, 2003), [(Bourguignon et al., 2008)]. Since then, it has been applied widely to answer a multiplicity of issues. More recently for the creation of the Global Income Distribution model in [(Bourguignon & Bussolo, 2013) ADD Busolo et al. 2010,

⁸Several packages are available to address the reweighting in practice. For instance, the MIKMOD-Est model in Germany (Flory & Stöwhase, 2012) relies on a package developed by Quinke, which bears similarities to Merz's "Adjust" reweighting algorithm (Merz, 1994). Additionally, the GREGWT algorithm, initially crafted by the Australian Bureau of Statistics (ABS) for spatial microsimulation purposes, is employed for survey reweighting to align with Australia-wide and capital city benchmarks. NATSEM utilizes the GREGWT algorithm in its SpatialMSM model (Tanton & Vidyattama, 2011).

Other alternatives include CALMAR, developed by INSEE in France (INSEE, 2011), and Clan97 (Anderson & Nordberg, 1998, as cited in Immervoll et al., 2005: 6). Notably, all three algorithms are implemented using SAS. Furthermore, Stata offers its own set of algorithms, such as CALIBRATE (D'Souza, 2011), Reweight (Pacífico, 2011, as mentioned in De Blander et al., 2013: 8; later revised as SReweight, Pacífico, 2014), and Gomulka (1992, as referenced in Brewer et al., 2011).

Buddelmeyer et al. (2012: 43) adopt the approach Cai (as cited) proposed in a two-step reweighting process to align with demographic totals and employment/unemployment levels, as provided by a CGE model. Lastly, combinatorial optimization (CO; Williamson et al., 1998) represents an alternative method that, to our knowledge, has yet to be utilized to "dynamize" a static model or dataset. CO entails selecting an appropriate combination of households from survey data to meet predefined benchmark constraints. For further insights and a comparison with GREGWT, refer to Tanton and Vidyattama (2011).

Table 2: Reweighting example. # restrictions, by number of population partitions

Population partition	#	Description
Age groups	a=3	{0-14,15-59,60+}
Sex groups	b=2	{female, male}
Schooling groups	c=3	{0-6, 7-12, 13+}
Urban residence	u=2	{rural, urban}
Labor status	f=2	{not in labor force/unemployed, employed}
Employment sectors	e=3	{agriculture, industry, services}
Microsimulation approach	# of restrictions	
Reweight on marginals	$[(3+2+3+2)+3]=13$	
Reweight on joint	$\underbrace{[(2x2x3x2) + (1x2x2)]}_{Dem.} + \underbrace{[(4x2x3)]}_{Emp.} = 13 + 24 = 37$	
Population partition	#	Description
Age groups	a=7	{0-9,10-19,...,50-59,60+}
Sex groups	b=2	{female, male}
Schooling groups	c=3	{0-6, 7-12, 13+}
Urban residence	u=3	{rural, urban}
Labor status	f=2	{not in labor force/unemployed, employed}
Employment sectors	e=3	{agriculture, industry, trad.services, modern services}
Microsimulation approach	# of restrictions	
Reweight on marginals	$[(7+2+3+2)+5]=19$	
Reweight on joint	$\underbrace{[(5x2x3x2) + (2x2x1x2)]}_{Dem.} + \underbrace{(5x2x3)}_{Emp.} = (60 = 8) + 30 = 98$	

2011, and 2012]]⁹. Applications to climate change policy are sparse ((World Bank, 2023) and (Group, 2023)). An application of macro-micro simulation, including a demand system to incorporate consumption changes as a behavioral reaction to energy price changes due to carbon taxes in Spain, is included in (Labandeira et al., 2009). Similar applications to carbon taxation in Mexico and in Ireland, but without a macro model on top, are performed by (Renner et al., 2018) and [(Tovar Reaños & Lynch, 2019)].

The *behavioral* approach refers to a microsimulation that combines a static aging approach to emulate changes in main demographic components (population numbers, composition by sex and age, schooling distribution, and urbanization) and a dynamic aging approach with rescaling to emulate changes in the employment and wage structure. A *behavioral* microsimulation usually has at least three steps:

1. alignment of baseline microdata with population projections to account for changes in the demographic structure, usually in joint age/sex/education-level forecasts and marginal urban/rural projections,
2. reallocation of workers and a estimation of counterfactual wages in order to align the baseline employment distribution with macro forecasts of counterfactual employment structures, usually by economic sector and workers' type,
3. rescaling of baseline wage distributions to align with forecasted wage bill growths, usually by economic sector (and, ideally, by workers' type),

Sometimes, a further alignment of simulated labor earnings to align it with macro forecasts of total income or consumption growth.

3.2.1 Aligning with future demographic structure

The first step in *behavioral* microsimulation is to align the baseline microdata with specific counterfactual pathways that characterize the evolution of the demographic structure. As in the reweighting approach, the behavioral approach calculates a new set of household importance weights to align the baseline survey population structure with a projected demographic structure for a given counterfactual scenario and target year. Yet, it leaves alignment with the counterfactual employment structure out of the static aging procedure. Formally, given a n -tuple that describes population partitions of a projected demographic structure in terms of age $\hat{a}_{st}$, sex, $\hat{b}_{st}$ schooling, $\hat{c}_{st}$ and place of residence,

$$\left[\hat{a}_{st} \cdot \hat{b}_{st} \cdot \hat{c}_{st}; \hat{u}_{st} \right]$$

so that

$$\hat{H}_{st} \cdot 1 \left[\hat{a}_{st} \cdot \hat{b}_{st} \cdot \hat{c}_{st}; \hat{u}_{st} \right] = \sum_h \sum_i \tilde{\lambda}_{ih,st} 1 \left[\hat{a}_{st} \cdot \hat{b}_{st} \cdot \hat{c}_{st}, \hat{u}_{st} \right] \quad (26)$$

where $1[\cdot]$ is the indicator function, and $\tilde{H}_{st}$ is the new total population.

As before, To find a distribution of household weights, $\lambda_{h,st}^*$ from the initial distribution of household weights $(\theta_{h,b0}^*)$ we solve the same optimization problem, imposing a set of constraints over populations partitions defined by the joint distributions of gender, age, and schooling and the marginal distributions of urbanization. The solution to the optimization problem will determine the growth rates of baseline importance weights for each population partition.

3.2.2 Sector reallocation

We now turn to the baseline microdata's alignment to the forecasted employment structure and labor earnings in the target year. The main objective of microsimulation is to forecast a future counterfactual welfare distribution that, 1) resembles - as closely as possible, what will happen with the income distribution at the target year in a given climate scenario, and 2) aligns with the macroeconomic aggregates imposed by external projections of demographic and macroeconomic aggregates. To achieve the latter, in the behavioral approach

⁹Global Income Distribution Dynamics (GIDD) discusses the link between economic growth and income distribution at the worldwide level and analyzes how income is distributed among individuals and households on a global scale, taking into account factors such as economic growth, inequality, and poverty: long-term time horizons and demographic changes in understanding growth and distribution dynamics. [Bussolo et al. (2009)]

the labor market model must be able to do two things: First, it needs to accurately predict who will move from non-employment or a shrinking sector to the expanding industries to align baseline labor supply, w_{b0}^{qr} , with a counterfactual labor supply that would be observed at a counterfactual scenario s at a target period t , l'_{b0}^{qr} . Second, it needs to be able to accurately predict what would the counterfactual labor earnings that a reallocated worker would earn and align the baseline labor earnings, w_{b0}^{qr} , with their macro level forecasts at a counterfactual scenario s at a target year t , w_{st}^{qr} . Not being able to choose the sample of workers that would be reallocated accurately or not assigning them with accurate counterfactual earnings that the reallocated workers would receive in their new sector may severely bias the microsimulation results.

Challenge 1: Predicting who moves where

The first challenge of the behavioral approach is to choose a (potentially large) set of workers moving out of non-employment and shrinking sectors into expanding sectors (usually industry and service sectors). Given a macro model forecast of the total number of workers who will end up in each economic sector, the behavioral approach will choose a particular set of individuals to be moved among an almost endless number of partitions that would satisfy the macro constraints. In particular, the behavioral approach estimates a set of parameters that describe the conditional probability of being a worker in a specific economic sector, given some individual characteristics, e.g., age, sex, and educational attainment.

It then ranks workers according to their probabilities (in reduced-form formulations) or implied (utilities) and applies variations of the above reallocation algorithms to identify the "most likely movers" - workers employed in a sector but have the lowest probability of (or derived the lowest utility from) being employed in that sector and to assign them from non-employment and shrinking sectors to the expanding sectors.

Under a behavioral approach, a microeconomic rule (model) is assumed (estimated) to select those workers who will increase their expected earnings if they "choose" to move to an expanding sector or those who will reduce their expected earnings if they are "forced" to leave a shrinking sector. While many microeconomic modeling alternatives support discrete choice among labor status and economic sector, the multinomial logit is the most widely used model for labor supply. The multinomial logit is based on a Random Utility approach that defines the selection of a given choice in terms of comparisons of absolute values of utility rather than by solving conditions that guarantee equality of marginal utilities across several variables of interest ¹⁰. The multinomial logit accommodates well to situations when attributes change only across observation cases but not across choices ¹¹. A caveat of this approach is that it assumes independence of irrelevant activities, i.e., that individuals face a single one-shoot choice between being non-employed or working in one of the R economic sectors. Because labor markets in developing countries are usually segmented and, therefore, not independent, the model has sometimes been replaced by one that assumes that the choices are sequential, with a participation decision first that is independent of the selection of economic activity and a subsequent choice of sectors that is conditional on the former ¹². Because of the challenges in modeling occupation, some authors had sometimes assumed simple random reallocation (?, ?).

Under the multinomial logit specification, the probability of observing an individual i working in a given sector, r , can be modeled as,

$$Pr_{ih,st}(r) = F(X_{ih}, Z_{.h}; \lambda_{.h,st}) = \frac{e^{\alpha_r \cdot X_{ih} + \beta_r \cdot Z_{.h}}}{e^{\alpha_r \cdot X_{ih} + \beta_r \cdot Z_{.h}} + \sum_{r \neq R} e^{\alpha_r \cdot X_{ih} + \beta_r \cdot Z_{.h}}} \quad , r \in \{0, 1, \dots, R\} \quad (27)$$

where X_{ih} and $Z_{.h}$ are vectors of individual and household characteristics for individual i in household h , respectively. Since labor supply decisions between household members are interdependent, it is useful to estimate different models, at least for household heads and other household members, where observed characteristics of one group are used as explanatory variables for the labor supply decision of the other. Model estimates assign a score to all people of working age based on their observed characteristics and new importance weights. Some simulations set the estimated utility as the sorting score - under certain assumptions about the disturbance distribution. Other, set the estimated probability as the sorting score.

¹⁰For more details, see [(McFadden, 1984)]. Although initially devised for explaining transportation choices, variations and extensions of this model have become a staple of labor supply studies. For a brief history of the evolution of labor supply models, see [(Aaberge & Colombino, 2014)].

¹¹Conditional logit approaches are more appropriate when attributes change across choices, and mixed logit approaches are more appropriate when characteristics change across choices and cases. For more details, see [(Cameron & Trivedi, 2005), pages 491-495].

¹²This approach would be compatible with a sequential probit estimation.

Assuming this model can be estimated satisfactorily, the alignment of simulation aggregates to macro forecast requires reallocating individuals in or out of employment and workers across sectors up to a point where both simulation aggregates and macro forecasts coincide.

To see the assumptions behind this approach, let's assume there are only two sectors, a *shrinking* sector, $r = r_0$, (say agriculture), and an expanding sector, $r = r_1$, (say industry and services). Furthermore, let's assume only two types of workers: unskilled workers, $q = q_0$, and skilled workers, $q = q_1$. Define the odds ratio between the proportion of type q_0 workers in the target scenario s at time t and the proportion of type q_0 in the baseline scenario at the initial year based on macroeconomic forecast given by

$$\frac{\hat{\eta}_{st}}{\hat{\eta}_{b0}} = \frac{\hat{l}_{st}^{r_1 q_0}}{\hat{l}_{st}^{r_1 q_1} + \hat{l}_{st}^{r_1 q_0}} / \frac{\hat{l}_{b0}^{r_1 q_0}}{\hat{l}_{b0}^{r_1 q_1} + \hat{l}_{b0}^{r_1 q_0}} \quad (28)$$

Similarly, let the simulation counterpart be given by,

$$\frac{\eta'_{st}}{\eta'_{b0}} = \frac{l'_{st}{}^{r_1 q_0}}{l'_{st}{}^{r_1 q_0} + l'_{st}{}^{r_1 q_1}} / \frac{l'_{b0}{}^{r_1 q_0}}{l'_{b0}{}^{r_1 q_0} + l'_{b0}{}^{r_1 q_1}} \quad (29)$$

If $\frac{\hat{\eta}_{st}}{\hat{\eta}_{b0}} > \frac{\eta'_{st}}{\eta'_{b0}} > 1$, once we account for the changes in the demographic structure ¹³, the supply of workers type q_0 in the expanding sector r_1 is more likely to occur in the macro forecast than in the simulation aggregates. We need to assign a fraction of these workers to the expanding sector to align odds ratios. Specifically, the number of workers type q_0 in the expanding sector r_1 that we need to reallocate to the expanding sector will be given by,

$$l^* = \eta'_{b0} \frac{\hat{\eta}_{st}}{\hat{\eta}_{b0}} (l'_{st}{}^{r_1 q_0} + l'_{st}{}^{r_1 q_1}) - l'_{st}{}^{r_1 q_0} \quad (30)$$

On the other hand, if $\frac{\hat{\eta}_{st}}{\hat{\eta}_{b0}} < \frac{\eta'_{st}}{\eta'_{b0}} < 1$, once we account for the changes in the demographic structure, the supply of workers type q_0 in the expanding sector r_1 is less likely to occur in the macro forecast than in the simulation aggregates. To correct for this overshooting of workers type q_0 at sector r_1 (probably caused by the alignment of the demographic changes), we will need to assign a fraction of these workers to the contracting sector.

Allowing workers of different types of skill to switch sectors is a big assumption for long-term micro-simulations because it assumes that unskilled and skilled workers are perfect substitutes, i.e., a unitary elasticity of substitution between unskilled and skilled labor. Furthermore, it also assumes that people - and their households, do not need to relocate to different regions to switch sectors. If the simulation exercise aims to analyze the welfare effects of mitigation policies - which are likely to reallocate low-skill workers from carbon-intensive industries in particular regions to high-skill jobs in carbon-neutral sectors in other regions, then the assumption might be controversial.

If we are willing to make these assumptions, reallocating from the shrinking sector to the expanding industry can be done differently. A common approach used in the literature follows Bourgingnon [2016] and assumes a waiting queue exists to enter the expanding sector. The waiting queue includes those in the contracting sector and those who are out of the labor force or unemployed. Workers in the queue are sorted according to a score usually defined in terms of the predicted utility or probability of being in the expanding industry. Most likely, movers will be those who are not in the expanding sector but have a higher *a priori*

¹³Let the equilibrium supply of workers of type q at economic sector r , in the baseline scenario at the initial year, and in the counterfactual scenario s at year t , respectively be given by,

$$l'_{b0}{}^{qr} = \sum_h \sum_{i \in h} \left[\sum_q \sum_r \lambda_{ih,b0}^{*qr} \right]$$

$$l'_{st}{}^{qr} = \sum_h \sum_{i \in h} \left[\sum_q \sum_r \tilde{\lambda}_{ih,st}^{*qr} \right]$$

Notice that, even without effective reallocation, changes in demography will affect $l'_{st}{}^{qr}$

probability of being employed in that sector. On the other hand, reducing employment in a contracting sector would imply a waiting queue to exit the contracting sector, with all potential candidates sorted such that most likely movers are those in the contracting sector but with the lowest *a priori* probability of being employed in that sector.

A simple procedure is incrementing all residuals in the occupation model by the same amount until we align simulation aggregates with macro forecasts. As Bourgingnon [2016] suggests, this reallocation procedure is equivalent to reweighing whenever the specification of the occupational model fails to select people into sectors conditionally based on observed individual and household characteristics. In practice, increasing the power of the occupation model requires: 1) being careful with the specification of the number and size of employment sectors, avoiding defining industries that account for a small fraction of employment (such as unemployment or specific two-digit sectors), and 2) consider the interdependency of occupation selection between the members of the same household, estimating first an occupation model for the household head, conditionally on the characteristics of all family members, and then for other household members conditionally on the decision of the household head.

Challenge 2: Assigning counterfactual labor earnings to reallocated workers

A second challenge for the behavioral approach is the assignment of counterfactual labor earnings to those who move from no-employment states or shrinking sectors to expanding economic sectors. This is usually done by estimating linear "Mincer" equations by employment sector. Mincer equations assign labor market returns to observed productive characteristics (usually, age/sex groups and schooling attainment) and unobserved characteristics (based on their rank). If an individual switches sectors, an income is calculated based on his observed productive characteristics, the returns observed in the destination sector, and his rank in the distribution of unobserved characteristics (estimated by the residual) if an individual moves into employment from unemployment or non-participation, a position in the rank in the distribution of unobserved characteristics is randomly drawn.

The wage-earnings equation is given by:

$$\log w'_{ih,b0}{}^{qr} = X_{ih,b0}^{qr} \beta^{qr} + \varepsilon_{ih,b0}^{qr} \quad (31)$$

where $w'_{ih,b0}{}^{qr}$ are the yearly expected labor earnings of individual i in household h at the baseline b in initial year t , $X_{ih,b0}^{qr}$ is a vector of productive characteristics that at least include the completed years of schooling and potential experience, and if possible also potential labor market segmentation effects across regions, firm sizes, races, or other relevant attributes. The vector β^{qr} is a vector of coefficients capturing the returns to productive characteristics at the baseline b , and $\varepsilon_{ih,b0}^{qr}$ represents the effects of unobservable characteristics (assumed to be orthogonal to $X_{ih,b0}^{qr}$). For the sake of simplicity, the residual distribution is supposed to be normally distributed with mean zero and sector and workers' type specific standard deviation $\hat{\sigma}_{b0}^{qr}$.

Therefore, the counterfactual wage distribution will be given by,

$$w'_{ih,b0} = \begin{cases} w_{ih,b0}^{qr} & \text{if } r = r' \\ \hat{w}_{ih,b0}^{qr} & \text{if } r \neq r' \end{cases}$$

where,

$$\hat{w}_{ih,b0}^{qr} = \begin{cases} X_{ih,b0}^{qr} \hat{\beta}^{qr} + \hat{\varepsilon}_{ih,b0}^{qr} \cdot \hat{\sigma}_{ih,b0}^{qr} & \text{for } e_{ih,b0} = 0 \rightarrow e_{ih,st} \neq 0 \\ X_{ih,b0}^{qr} \hat{\beta}^{qr} + \hat{\varepsilon}_{ih,b0}^{qr} \cdot \frac{\hat{\sigma}_{ih,b0}^{qr}}{\hat{\sigma}_{ih,b0}^{qr}} & \text{for } e_{ih,b0} \neq 0 \rightarrow e_{ih,st} \neq 0 \end{cases}$$

and $\hat{\varepsilon}_{ih,b0}^{qr}$ is a random draw from a standard normal distribution, $\hat{\varepsilon}_{ih,b0}^{qr}$ is the predicted disturbance for those with labor earnings (all employed workers at the baseline), and $\hat{\sigma}_{ih,b0}^{qr}$ is the estimated variance of the error term the earnings equation for sector q .

3.2.3 Alignment with changes in the wage structure (by sector and workers' type)

With both a counterfactual employment choice and counterfactual labor earnings (for those who move into work or change sector of employment), the next step is to adjust the returns to labor by worker's type, q ,

and industry, r , to match the labor productivity growth forecasts coming from the macroeconomic model. As before, this is done by calculating residual growth rescaling factors by sector and industry that allow the simulated counterfactual labor earnings to match labor productivity growth forecasts, ensuring consistency between the macro and micro outcomes.

While in the reweighting approach, the new set of importance weights accounted for changes in both demographic and employment structure, in the behavioral approach, the new set of importance weights accounts only for changes in the demographic structure. Reallocation effects, which originated from the changing size of sectors and the associated job reallocation of different types of workers, are accounted for by the counterfactual distribution of employment and related labor earnings. To align baseline labor earnings by workers' type, q and industry, r , with the counterfactual forecasts of the macroeconomic model, we calculate a *residual* growth in labor productivity, $\delta'_{W_{st}}{}^{qr}$, that guarantee that, for each workers' type, q , and each industry, r , the simulated average wage growth, $g'_{st}{}^{qr}$, aligns with the forecasted macro growth $\hat{g}_{W_{st}}{}^{qr}$. As before, the apostrophe in the variables denotes counterfactual distributions and the "hats" macro forecasts.

Let $\hat{g}_{W_{st}}{}^{qr}$ be the forecast for the overall labor productivity growth for each workers' type, q , and each industry, r , between counterfactual scenario s at target year t and the baseline scenario b at the initial period 0. Let $\bar{w}_{st}{}^{qr}$ and $\bar{w}_{b0}{}^{qr}$ be the average labor earnings for counterfactual scenario s and period t , and the average labor earnings at the baseline scenario b and base year 0.

$$\bar{w}_{b0}{}^{qr} = \frac{\sum_i \sum_{i \in h} (l'_{ih,b0}{}^{qr} \cdot w'_{ih,b0}{}^{qr} \cdot \lambda_{ih,b0}^*)}{\sum_h \sum_{i \in h} (l'_{ih,b0}{}^{qr} \cdot \lambda_{ih,b0}^*)} \quad (32)$$

$$\bar{w}_{st}{}^{qr} = \frac{\sum_h \sum_{i \in h} (l'_{ih,b0}{}^{qr} \cdot w'_{ih,b0}{}^{qr} \cdot \lambda_{ih,st}^*)}{\sum_h \sum_{i \in h} (l'_{ih,st}{}^{qr} \cdot \lambda_{ih,st}^*)} \quad (33)$$

Notice that, while in the reweighting approach the new weights were the only force at play changing the simulated wages and their averages, $\bar{w}_{st}{}^{qr}$, in the behavioral approach there are three sources of change of the simulated wage and their averages, $\bar{w}_{st}{}^{qr}$: 1) the new weights due to changes in the composition of the labor force due to demographic change, 2) the new labor force employment distribution of employment status and sector of employment, and 3) their associated counterfactual labor earnings distribution - with counterfactual wages for those who move into the labor force and those who change sectors of employment. Of course, it is unlikely that all these movements will be enough to align microsimulations with macro forecasts, so - as was the case of the reweighting approach, there is a need to adjust for any residual growth in labor productivity. The residual growth in labor productivity will be implicitly defined by,

$$\hat{g}'_{W_{st}}{}^{qr} = \delta'_{W_{st}}{}^{qr} \cdot \bar{w}_{st}{}^{qr} / \bar{w}_{b0}{}^{qr} \quad (34)$$

As with reweighting, the behavioral simulations do not adjust baseline wages using the overall labor productivity growth forecasts, $\hat{g}_{W_{st}}{}^{qr}$, but only a residual productivity growth, $\delta'_{W_{st}}{}^{qr}$, that accounts for the part of labor productivity growth not accounted for by the growth in labor productivity growth due to changes in the demographic and employment composition of the labor force.

3.2.4 Alignment to overall economic growth

As with reweighting, there may be a need to adjust simulations further to be consistent with aggregate forecasted growth of total per capita income (or per capita private consumption). To align simulated growth rates in per-capita income with the macro forecasts, a residual **per-capita income rescaling**, $\gamma_{Y_{st}}$, that ensures that the simulated growth rate of the average counterfactual household income in the survey is aligned with its forecasted counterpart for the macro model is implicitly defined as,

$$\hat{g}_{Y_{st}} = \gamma_{Y_{st}} \cdot \bar{y}_{st} / \bar{y}_{b0} \quad (35)$$

where $\hat{g}_{Y_{st}} = \bar{y}_{b0} / \bar{y}_{st}$ is the macro forecast of per capita income (or consumption) growth between scenario s at time t and the baseline scenario b at initial year 0, and $\bar{y}_{b0}$ and $\bar{y}_{st}$ are, respectively, baseline per capita income in the baseline scenario b at initial year 0, and the simulated percapita income in counterfactual scenario s at the target year t

While implementation can be more complex, the behavioral method offers a theoretical foundation and a granularity of results that is not available in the reweighting approach. On the other hand, in addition to modeling complexity, there are other substantive caveats. First, the behavioral responses are based on individual preferences for consumption and leisure, as manifested in the currently available survey. It isn't easy to accept that these observations will represent the preferences of these individuals in the medium term or of other individuals in the long term. Hence, simulations of behavior and economic choice may seem implausible.

In particular, if the final objective is to accurately predict the counterfactual distribution of employment and labor earnings that would be observed under scenario s in future period t , it is essential to note a few important caveats.

First, even when the behavioral model is depicted as a simultaneous equation model with a full dependence structure, a piecewise estimation procedure that assumes independence between model blocks is imposed in practice. In other words, not only did we assume that the allocation of individuals across occupations occurs independently from the distribution of random terms of the earnings equations - which contradicts utility maximization, but also that there is no correlation between the labor earnings equations across different sectors (i.e., no selection into employment effects)¹⁴. Second, even if the earnings model was correctly specified, *Mincer* linear equations are more appropriate for estimating mean responses, not distributional sensitive ones. Alternative non-parametric approaches that partition the employed population according to observable characteristics and assign a new labor income to individuals who move from non-employment to employment or switch between employment sectors according to their corresponding partitions might improve the estimation of counterfactual distributional in the absence of self-selection into economic sectors. Finally, because we do not observe labor earnings for non-employed people at the baseline, the estimation of counterfactual labor earnings for those who move into employment has to rely on a random draw value from a standard normal distribution whose value is adjusted to the dispersion of the destination sector. Given that most labor earnings equations explain at most 2/3 of the overall variations in earnings, this means that, at least for those who move from non-employment, the counterfactual wage relies mainly on a random draw.

3.3 Incorporating price effects

Suppose macroeconomic models include forecasts of the distribution of price changes among households. In that case, it may be interesting to account for the effects of agricultural and non-agricultural prices and energy and non-energy prices.¹⁵ In those cases, it is helpful to distinguish between *real*, $Y_{h,st}^r$, and *nominal* household total income, $Y_{h,st}$. Real values are obtained by dividing nominal values by a household-specific price deflator, $P_{h,st}(\hat{\phi}_{h,st}, \bar{P}_{st}^f, \bar{P}_{st}^n)$. The household-specific price deflator depends on *average* food and non-food price indexes for each counterfactual scenario s at each particular year t and on a household-specific forecast of the share of the budget spent on food. Formally,

$$Y_{h,st}^r = \frac{Y_{h,st}}{\bar{P}_{h,st}(\hat{\phi}_{h,st}, \bar{P}_{st}^f, \bar{P}_{st}^n)} = \frac{Y_{h,st}}{\hat{\phi}_{h,st} \cdot \bar{P}_{st}^f + (1 - \hat{\phi}_{h,st}) \cdot \bar{P}_{st}^n} \quad (36)$$

Equation 36 captures the dual effect of price changes. On the one hand, price variation will change the household-specific share of food. On the other hand, price increases will be associated with a loss of purchasing power. Typically, macroeconomic models normalizes initial consumer prices equal to 1, $\bar{P}_{b0}^f = \bar{P}_{b0}^n = 1$ so that, $Y_{h,b0}^r = Y_{h,b0}$.

where $\bar{P}_{st}^f$ and $\bar{P}_{st}^n$ are food and non-food price indices and $\hat{\phi}_{h,st}$ is the *forecasted* proportion of household's h budget spent on food. This parameter can be obtained by estimating the Engel curve that relates budget shares with the level of per capita household income with baseline data using the following specification,

$$\phi_{h,b0} = \beta_0 + \beta_1 \ln(Y_{h,b0}) + e_{h,b0} \quad (37)$$

¹⁴Furthermore, suppose the occupational choice model's random component is independent of the earnings equations' random components. Since households usually have more than one member, and there is likely to be a correlation between occupational choice and earnings across members, this is not usually the case.

¹⁵This is usually available for most CGE-type models. In this section we follow the approach proposed by Osorio Rodarte (2016)

where $e_{h,b0}$ is a vector of household-specific errors that are assumed to be distributed with $E(e_{h,b0}) = 0$ and $V(e_{h,b0}) = \sigma^2$. Assuming that both, estimated parameters, $\hat{\beta}_0$ and $\hat{\beta}_1$, and estimated distribution of household-specific errors remain constant over scenarios and periods, the simulated budget share spent on food for household h , $\hat{\phi}_{h,st}$, at each counterfactual scenario s at year t , will be given by,

$$\hat{\phi}_{h,st} = \hat{\beta}_0 + \hat{\beta}_1 \ln(\hat{y}_{h,st} \cdot Y_{h,b0}) + \hat{e}_{h,b0} \quad (38)$$

where $\hat{y}_{h,st} = y_{h,st}/y_{h,b0}$ is the household specific income growth between the counterfactual scenario s and the baseline scenario s for year t . The parameter $\hat{\beta}_1$ approximates the changes in food shares brought about by a change in per capita income.

4 The "shockwaves" approach

In the World Bank report *Shock Waves: Managing the Impacts of Climate Change on Poverty* [(?, ?)], accompanying methodological paper [(Hallegatte & Rozenberg, 2017)] and subsequent updated estimates [(Hallegatte, Fay, & Barbier, 2018)] the authors criticize aggregate (country-level) assessments of the climate change impact focused in GDP losses. Using country-level GDP losses as the metric to measure the country's social cost of climate change does not adequately incorporate all distributional impacts of how climate change will affect the poorest of the poor, and, therefore, they do not account for the actual value of adaptation and mitigation mechanisms. Poor households are more exposed and vulnerable to climate hazards, bearing a substantial burden of the associated damages (Mendelsohn et al., 2000; Tol, 2002, 2009; Hope, 2006; Mendelsohn et al., 2006; Stern, 2006; Nordhaus, 2014; Hsiang et al., 2017). Since conventional Top-Down forecasts of the household's welfare distribution only account for changes in the employment and wage structure induced by aggregate- (macro-) level climate damages, they fail to account for the heterogeneous distribution of impacts across households and, especially the disproportionately large climate impacts on the poor¹⁶. As (Hallegatte, Fay, & Barbier, 2018) put it, "investigating the impact of climate change on welfare requires a different approach, focused on people who play a minor role in aggregate economic figures, are often living within the margins of basic subsistence in rural areas (Castañeda et al., 2018)".

To adequately capture the distributional impact of climate change at the household level, and not only downstream GDP growth losses through the labor market, the authors propose a "Shockwaves" approach¹⁷. The *Shockwaves* can be defined as a Monte Carlo microsimulation (MCM). Generally, MMCs define not only a mathematical model that relates key parameters to specific outcomes but also the probability distribution of those key parameters. With such probability distributions, MCMs can generate and compare an extensive collection of counterfactual outcomes - associated with a wide range of counterfactual scenarios instead of concentrating on one and only one set of parameters associated with a given counterfactual scenario. The extensive collection of outcomes is usually obtained via repeated random sampling of particular parameter draws. It is essential to notice that a MMC will be informative when the extensive collection of simulated outcomes describes *the distribution of potential results*. That is, not only *the range* of potential outcomes, but also *the likelihood* of each of those potential outcomes. For MCM to be able to produce a distribution of potential outcomes, the key is not only to define a comprehensive model with parameters that account for the critical features of a system but also to adequately define the probability distribution of those parameters, which will define the likelihood distribution of the simulated outcomes.

The main contribution of the shockwaves approach is that it builds in the flexibility of MCM simulations to incorporate a variety of micro-level climate damages (on crop yields, the distribution of stunting, malaria and diarrhea, heat stress, and natural disasters) to explore a distributional effect of climate change not accounted

¹⁶In the same way, global GDP climate losses miss the disproportionately large climate effects on particular countries and regions.

¹⁷The *Shockwaves* approach was the product of the research program on "Poverty and climate change" at the World Bank, which led to the publication of the report *Shock Waves: Managing the Impacts of Climate Change on Poverty*. The main results are that the range of the number of people falling into poverty due to climate change is between 32 million and 132 million in most scenarios. Socioeconomic drivers play a significant role: optimistic baseline scenarios (rapid and inclusive growth with universal access to essential services in 2030) halve poverty impacts compared with the pessimistic baselines. Health impacts (malaria, diarrhea, and stunting) and the effect of food prices are responsible for most of the impact. The effect of food prices is the most critical factor in Sub-Saharan Africa, while health effects, natural disasters, and food prices are all important in South Asia.

for the conventional Top-Down simulations. In particular, [(Hallegatte & Rozenberg, 2017)] and [(Hallegatte, Fay, & Barbier, 2018)] build thousands of scenarios incorporating micro-level damages to explore the range of extreme poverty numbers that can be expected in 2030 when climate-related hazards are introduced not at the macro-level but at the household-level, directly affecting household's income generating capacities and returns. In particular, the authors explore climate-related impacts on crop yields (that account for changes in food prices), climate-related health impacts (malaria, diarrhea, and stunting), heat stress effects, and the destruction of assets (and their income effects) due to natural disasters ¹⁸.

Because there is intrinsic uncertainty on the development pathways, including demographic and growth projections, they use the flexibility of MCM to vary many of the key parameters and produce a wide range of possible outcomes. In this sense, the shockwaves take the simulation of a particular counterfactual scenario as only one roll -or potential drawn, from an infinite set of possible outcomes generated by an underlying distribution, which may or may not be observed in the future. For the authors, this modeling approach helps incorporate uncertainty along many dimensions, including micro-level damages and demographic and growth projections.

Precisely because the Shockwaves approach produces thousands of microsimulations, where each simulation is associated with a different income distribution and aggregate poverty measures, it breaks free from macro to micro-alignment. Conventional "Top-Down" downstream changes in the employment and wage structure induced by aggregate economic growth and changes in the demographic structure to the household level, making sure the new structure is consistent and aligns micro-changes to the macro-changes. Because the *Shockwaves* assumes each parameter draw is independent of the others, it explicitly forgone macro consistency and alignment.

The *Shockwaves* approach generates a large collection of counterfactual outcomes using a MCM that follows at least three steps:

1. **Define the modeling approach**, in this particular case, define the household per-capita income as a function of key parameters.
2. **Define the probability distribution of the key parameter**,
3. **Perform a large number of microsimulations**, using a MCM procedure.

Following this approach, in the rest of this section, we describe the baseline data transformation that constructs synthetic observations to ease the MCM procedure, the incorporation of micro-level damages into the household income-generating equation, the definition of the probability distribution for key simulation parameters, and, finally, the mechanics of the Shockwaves algorithm.

4.1 Synthetic observations

As we described in the last section, a vital part of a microsimulation approach to analyzing the welfare effects of climate change is choosing a baseline survey with enough quantity and quality of information. Sample size considerations are crucial for reweighting methods -changes in the importance weights to account for demographic and employment changes, especially at a very disaggregated level. Quality information on employment, particularly annual labor earnings by sector, is the primary source of future per capita income distribution changes. Unfortunately, not all surveys have size large enough and consistent labor earnings information available. Most Sub-Saharan African countries, in particular, and informal developing economies, in general, have surveys where labor earnings variables have problems of non-response, out-of-range values, and significant measurement errors.

Although it is not crucial for the approach, it is worth mentioning that the Shockwaves approach uses "synthetic" observations instead of individual observations. Synthetic observations collapse the information of individual observations into a specific number of groups to produce a parsimonious representation of all households that can reproduce the population structure of the baseline data. To do this, the shockwaves approach rewrites the total household income of a given household, Y_h , as,

$$Y_h = \sum_r \hat{w}_p n_{h,p} + z_h, \quad \text{for } p \in \{4, 5, 6, 7, 8, 9\} \quad (39)$$

¹⁸In addition, the approach also assesses the benefits and limitations of introducing redistribution mechanisms (through pensions and social protection).

Where $\hat{w}_{hr}$ can be interpreted as the (estimated) household average per-capita labor income coming from $n_{h,p}$ household members in partition p and h is household specific residual. Population partitions are defined as a combination of three age groups, two skill levels, and four labor market sectors (the non-employment status and the three primary economic sectors) for a total of only nine population partitions:

- children (up to 14 years old) ($j = 0$)
- elderly (65+ year old)
- adult non-employed
- adults employed in agriculture (by two skill levels)
- adults employed in industry (by two skill levels)
- adults employed in services (by two skill levels)

Several points must be noted regarding this approach. For countries with only consumption data or pernicious problems of non-response or out-of-range values in labor earnings, synthetic observations construct a proxy by assigning a (household-specific -?-) equal share of consumption to all household members of a given partition. For countries with labor earnings information, synthetic observations can amiorate the problem of significant measurement errors, especially when dealing with short reference periods. A second advantage of using only nine synthetic observations (i.e., only nine reweighting partitions) is that it facilitates the estimation of counterfactual importance weights that allow the projection of baseline survey data into specific future counterfactual scenarios. Of course, the gains in simplicity come at the cost of leaving aside much granularity from labor earnings variations within groups from finer demographics (e.g., distinctions by male and female) or finer sectors of employment classifications (e.g., formal and informal jobs).

4.2 Incorporating micro-level climate induced damages

A crucial distinction between the transmission mechanisms of the conventional "Top-Down" approach and the Shockwaves approach is that the latter incorporates micro-level climate-induced damages while the first only incorporates macro-level climate damages. More specifically, the conventional Top-Down approach to analyze the household's welfare impacts of climate change uses macro-level damages function, i.e., estimates of the damage of a given set of climate hazards on the aggregate GDP level to forecast aggregate growth effects and sectoral changes in wages and employment. It does not incorporate individual-level (micro-level) damages induced by climate hazards. Yet, it is clear that specific regions and households within those regions will bear a higher impact on climate hazards. Significant variations in the vulnerability and exposure to climate damage might exist that are not accounted for when the simulation considers only aggregate damages at the GDP level.

To incorporate the distributional effect of micro-level damages, the *Shockwaves* approach includes four climate-related damage mechanisms: 1) changes in agricultural yields (net of impacts on food prices), 2) health impacts on working hours (on stunting, malaria and diarrhea), 3) heat stress effects on labor productivity effect, and 4) natural disasters effects on household income due to the destruction of capital stocks. In most cases, the calibration of household-specific damages follows a stratified random sample approach. First, the population is divided into homogeneous groups (i.e., strata) according to particular characteristics observable in the simulation, say, their future level of per-capita income. Then, random samples of households and individuals within those households are selected to match a forecast of the future prevalence of a given hazard, e.g., Malaria's prevalence in the particular population strata. It is important to notice that this stratified random sample approach attempts to be consistent with current inequalities in exposure to a given hazard and to project the evolution of those inequalities into the future. It is also important to note that in most cases, household-specific damage is uniform within strata and independent of other socioeconomic characteristics such as demographics, skill, or employment sector. In reality, both individual exposure and vulnerability will likely depend on other socioeconomic variables. For example, the prevalence of Malaria is likely to depend not only on specific types of Malaria prevalent in specific regions but also on the health status of the population living in that area and, more importantly, on the availability of health care and

appropriate treatments. Table 3 describes the particular combination of exposures and damages used by Hallegatte et al. (2015) to calibrate micro-level damages.

More formally, let $\tilde{d}_k$ be the household-specific damage due to a climate hazard, k , the income generating equation of the shockwaves approach will incorporate micro-level damages as an individual-specific penalty that will result in income loss for both, specific households, in particular, and for the economy as a whole, in general.

$$\begin{aligned}
\tilde{Y}_{st}^S &= \sum_h \sum_{i \in h} [1 + \tilde{d}_{st,k}] \cdot y_{ih,st} \cdot [\tilde{\theta}_{.h,st}^*] \\
&= \sum_h \sum_{i \in h} [[1 + \tilde{d}_{st,k}] \cdot l_{ih,st} \cdot w_{ih,st} + z_{ih,st}] \cdot [\tilde{\theta}_{ih,st}^*] \\
&= \sum_h \sum_{i \in h} \left\{ \left[\sum_{q=1}^Q \sum_{r=1}^R [1 + \tilde{d}_{st,k}] \cdot l_{ih,b0}^{qr} \cdot w_{ih,b0}^{qr} \cdot \tilde{g}_{st}^{qr} \right] + [\tilde{g}_{Yst} \cdot z_{ih,b0}] \right\} \cdot [\tilde{\theta}_{.h,st}^*]
\end{aligned} \tag{40}$$

Three additional methodological points are worth noticing: New importance weights and rescaling factors are treated as draws of a known distribution. In other words, each simulation is tied to a given set of parameters defining demographic changes (in age and skill), employment changes, and labor productivity growth. The shockwaves simulation approach has specific choices regarding the definition of population partitions, the reweighting algorithms used, and the calibration of rescaling parameters; the simulation mechanisms are similar to those used by the reweighting approach. That is, it combines a static aging approach to simulate changes in demographic components and changes in the employment structure and rescaling procedures to simulate changes in the labor productivity and climate damage effects on either the utilization or remuneration of the productive capacity of the workers. It is essential to notice that because key parameters are drawn independently from the assumed distributions, the shockwaves are an unbounded simulation with no explicit alignment to any macroeconomic projection.

4.3 Defining a *range* for key parameters

Instead of choosing a particular development pathway defined by precise population projections (in terms of age-sex composition, skill accumulation, and urbanization levels) and macroeconomic growth forecasts (associated with particular employment and wage structure structural changes), the Shockwaves defines a *range* of plausible values for critical parameters and assumes almost all of them come from a uniform distribution within that range. Table3 present the selected scenarios or bounds (minimum and maximum values) used by Hallegatte et al. (2015). For demographic changes, scenarios were chosen based on population data (total population by age, sex, and education) corresponding to IIASA's SSP4 and SSP5. Bounds for structural change employment effects are based on the initial economic structure of each country and projected pathways, using the minimum and maximum change observed in historical data over the last 20 years. Bounds for participation rates also look at available historical data -for all countries and all periods, choosing slightly larger boundaries than the lowest and highest rates of change over 20 years. Finally, bounds for labor productivity growth use different productivity growth rates for unskilled workers in each economic sector and add a skill premium for skilled workers. Bounds for the labor productivity growth rate range add GDP growth and the working-age population growth from the SSP scenarios. Bounds for the skill premium use a range of 1 to 5 (as the current observed range across countries). In all cases, the parameters distribution is assumed to be uniform.

4.4 The MCM algorithm

Once a modeling structure has been specified, including, 1) a baseline data, 2) a household income-generating equation (as a function of key parameters), and 3) a distribution of key parameter values, the Shockwaves model can produce a large number of counterfactual outcomes using a MCM procedure. The MCM procedure, in essence, has three steps:

Table 3: A summary of micro-level damage assumptions

	Mechanisms					
1) Crop yields change (%) EAP ECA LAC MNA SAS SSA		Optimistic		Pessimistic		
		SSP4	SSP5	SSP4	SSP5	
		-0.74	-0.75	-3.2	-3.0	
		2.66	12.00	0.9	4.7	
		-0.21	-0.18	4.7	2.2	
		4.25	1.70	0.5	0.9	
		-1.23	-1.34	-4.5	-3.5	
		-1.39	-1.33	-6.1	-5.8	
2) Food prices change (%) EAP ECA LAC MNA SAS SSA		Optimistic		Pessimistic		
		SSP4	SSP5	SSP4	SSP5	
		0.37	-0.13	3.4	1.9	
		-2.62	-3.27	-0.1	-1.0	
		0.39	-0.01	-1.0	2.0	
		-2.03	-1.49	2.7	-0.8	
		3.29	1.46	7.7	4.9	
		0.74	0.23	7.1	3.1	
3) Health Stunting Damage (penalty % earn.) Malaria Occurrences (times/year) Damage (days out of work) Cost of treatment (US\$ PPP) Diarreah Occurrences (times/year) Damage (days out of work) Cost of treatment (US\$ PPP)	Cut-point (<i>c</i>) 8,000	Optimistic below <i>c</i> above <i>c</i>		Pessimistic below <i>c</i> above <i>c</i>		
		5.0		15.0		
	8,000	0.1		2.0		
		1.0		5.0		
		0.7		6.0		
	15,600	1.0		8.0		
		3.0		7.0		
		2.0		4.0		
	4) Heath-stress Agriculture Fraction (%) Damage (% earn.) Industry Fraction (%) Damage (% earn.) Services Fraction (%) Damage (% earn.)	Cut-point (<i>c</i>)	Optimistic below <i>c</i> above <i>c</i>		Pessimistic below <i>c</i> above <i>c</i>	
		10,000	0.8	0.80	0.8	0.80
		1.0		3.0		
10,000		0.5	0.10	0.5	0.10	
		1.0		3.0		
10,000		0.3	0.05	0.3	0.05	
		1.0		3.0		
5) Natural Disasters Damage (% earn.)		Cut-point	Optimistic Poor Non-poor		Pessimistic Poor Non-poor	
		-20	-10	-30	-15	

Source: Author's summary based on Hallegatte et al. (2015)

1. **Obtain a random draw** for each sensitive parameter that affects the household income-generating potential. Draws are obtained from a uniform distribution within predetermined bounds or a standard deviation from the most likely estimate.
2. **Adjust the households' sampling weights** to simulate changes in the demographic and (best-guess estimates of) changes in the employment structure. Changes in the demographic structure include aging shifts to account for changes in the marginal distribution of only three cohorts (children, people of working age, and elders), two schooling groups (skill and unskill), and four economic sectors (non-employment, agriculture, industry, and services). It does not include changes in the population by sex or urbanization patterns.
3. **Simulate counterfactual wage distributions**, by rescaling baseline distribution to account not only for sectoral changes in labor productivity but also climate damage impacts on either the utilization or remuneration of the productive capacity of the workers. Again, it is important to notice that rescaling is done without macro alignment. For example, rescaling to account for labor productivity growth does not account for the productivity gains coming from changes in the skill composition of the labor force. Also, the aggregation of household-level damages does not necessarily match with forecasted macro-level damage at the GDP level,

Before moving to compare the Top-Down and Shockwave approaches, a couple of points are worth noticing. In the first two implementations of the Shockwaves approach Hallegatte and Rozenberg (2017), the microsimulation of future scenarios breaks utterly free from alignment to future macro pathways using Monte Carlo simulations to vary assumptions of multiple/all parameters and thus model a range of possible outcomes. In contrast, the application of the Shockwaves approach in Peru uses macro linkage estimates as anchors (means) of a still large and ad-hoc range of possible future values.

5 A comparison of advantages and disadvantages

5.1 *Re-weighting* vs. *Behavioral "Top-Down"* approaches

As the comparison of the *behavioral* approach, in equation , and the *reweighting* approach, in equation reveals, the main difference between both approaches is in the alignment mechanisms used to impose forecasted aggregates in target years into the microdata simulations regarding two key variables: 1) the distribution of employment opportunities across individuals, and 2) their associated distribution of labor earnings.

On the one hand, alignment in the *behavioral* approach relies on the counterfactual distribution of wages at the baseline, $w_{ih,b0}$ caused by a counterfactual distribution of employment opportunities $l'_{ih,b0}$. Both, counterfactual employment opportunities and counterfactual wages use dynamic aging to assign some individuals to new jobs (status and sectors) and to calculate the "what-if" labor earnings they would have earned in the new sector. Static aging (updating weights), $\theta^*_{ih\tau}$ are only used to update baseline microdata to the new population pathway observed in the target year.

On the other hand, the *reweighting* approach does not need to construct either counterfactual employment allocations or counterfactual wages. All it needs to do is use static aging - finding a new set of household weights, $[\theta^*_{ih\tau} \cdot \lambda^*_{ih\tau}]$ that not only updates baseline microdata to reproduce the population pathways observed in the target year but also to match the forecasted employment structure. There is no need to assign counterfactual labor earnings to individuals because no individuals change their labor status, employment sector, or baseline wage. The *reweighting* approach is more straightforward because it does not require complex modeling. It simply assumes all families and members face a similar *prior* probability of remaining/becoming (inactive) and of changing employment (although the *posterior* probability depends on the reweighting algorithm). In other words, it assumes random job reallocation incorporates marginal employment numbers by status and sector as constraints in the reweighting procedure. Because it does not change workers sector it does not require counterfactual labor earnings estimation.

In theory, when well specified, behavioral models can elegantly characterize household members' opportunity sets and infer their preferences over occupational choices, i.e., to identify which members (e.g., household heads or young adults) are more likely to decide to remain/become (in)active or to change employment sector. If reallocation is selective, then transition probabilities will be conditional on joint distributions of individual

characteristics, status, and employment sector. In practice, this requires more than complex modeling techniques and quantity and quality data but several simplifying assumptions that are only sometimes true. More likely, behavioral models are best characterized as dynamic aging prediction rules that assign future labor status, economic sector, and counterfactual earnings to individuals to align simulations with macro forecasts. For behavioral models to be superior to other alternatives, their statistical models used to predict who moves where should be more accurate than a simple random reallocation or more uncomplicated sorting strategies. Out-of-sample, multinomial statistical approaches' accuracy in classifying working-age individuals across different labor market statuses is usually better than random guessing. Still, the same is only sometimes valid when the objective is to classify employed individuals into conventional expanding sectors such as industry and services. Whether the labor supply decision and sector allocation model can be estimated satisfactorily to make selective reallocation empirically different from random reallocation is an empirical question that depends on the quantity and quality of information available and the characteristics of the particular context under analysis. Figure?? illustrates the situation where even with models that predict significantly different probabilities for each sector of employment (Panel a), there is no difference between the baseline wage density of workers and the target wage density of the most likely movers according to their probabilities (Panel b).

The same is true for the models used to assign counterfactual earnings for those who move from non-employment to employment or to those who switch sectors. While the linear Mincer approach described here is widely used and popular, it is prone to self-selection problems and needs to be more accurate, even with good quality and quantity data. In most applications, within sample R^2 - the percentage of the total variance in labor earnings accounted for the model is not above 0.40, implying that most of the counterfactual labor income assignment is based on a rank-preserving imputation of prediction errors - which are assumed to be random for those who move from out-of employment into employment. Figure?? illustrates how large unexplain residuals can be (Panel a), and how counterfactual wages may collapse an unequal distribution of wages into a more condense one(Panel b).

On the other hand, the *reweighting* approach depends on the quantity of data, i.e., the distribution of sample sizes between partitions of worker types and employment sectors and the specific distribution of demographic characteristics and employment status and industry in the sample. As we will discuss, the main challenge of the *reweighting* approach is related to practical implementation choices such as the optimal population partition, to reweight over joint or marginal distributions of characteristics, or the selection of the reweighting algorithm.

Finally, well-defined *behavioral* models are more likely to be appropriate for **short-term** forecast horizons than for **long-term** forecast horizons, particularly for target years in the distant future ($t_0 + 30$) years from the baseline, but validation exercises may be needed to define which one performs better in predicting redistribution effects and under which circumstances. Different approaches may have different results depending on the quantity and quality of the available data and the country's particular circumstances (e.g., whether the baseline data has well-behaved labor earnings data available or relies on imputed labor earnings values based on some specific procedure). So, the choice between one approach or the other is context-dependent.

5.2 "Top-Down" vs "Shockwaves" approaches

The main difference between the "Top-Down" approach and the "Shockwaves" approach can be pinned down to one central issue: the difference in goals of the simulation exercise.

Shockwaves vs. "Top-Down" objectives

Both "Top-Down" approaches to microsimulation, the *behavioral* and the *full reweighting*, are interested in assessing the most likely effect of climate change on a given measure of household welfare. To do this, they compare a forecasted distribution of per-capita income in a baseline scenario, $s = 0$, with the forecasted distribution of per-capita income in another -more optimistic or pessimistic climate scenario. Both the baseline and the more optimistic or pessimistic climate scenario share a common demographic pathway and familiar economic growth drivers and transmission mechanisms (associated with employment and wage effects of structural change effects) so that the differences in a measure of aggregated welfare in a particular year, $t + 10, t + 20, t + 30$ periods from now, can be entirely attributed to the difference in climate-related hazards

and their damages (at the macro-level). On the other hand, the "Shockwaves" approach is interested in forecasting *a whole range* of potential values of a given household welfare measure, the extreme poverty headcount. In their view, the crucial goal is not to explicitly predict the single-best guess estimate of the future consequences of climate change but instead, to identify the extremes that define the range, i.e., more optimistic and the more pessimistic scenarios, and to look at "for the necessary conditions (or drivers)," i.e., the combination of random draws that lead to such scenarios.

Treatment of uncertainty

The differences in goals between the "Top-Down" approach and the "Shockwaves" approach are intrinsically related to different notions of uncertainties regarding the potential effects of climate change on household welfare. The "Top-Down" view assumes that the system model and its inputs can be described probabilistically and that a few alternative futures can be predicted well enough and to which probabilities can be assigned. At the other extreme, the "Shockwaves" are more closely related to the deepest level of recognized uncertainty, where we are only able to "bound" the future around many plausible futures not only because we lack the knowledge or data about the mechanism or functional relationships to analyze the welfare effects, but also because the potential for unpredictable and surprising events. Therefore, factors that can be seen as limitations under the perspective of one of the approaches (e.g., the disregard for macroeconomic consistency, the lack of probabilities or likelihood assigned to a large number of scenarios, and even the disregard for well-defined micro-level damage mechanisms), are seen as strengths under the alternative approach (Marchau, Walker, Bloemen, & Popper, 2019).

The challenge of the GIDD approach is to define where and how to incorporate uncertainties in long-term forecasts; the challenge of the *Shockwaves* approach is to determine where and how to reduce uncertainties in the simulation of alternative scenarios, or at least how to condition the likelihood of observing a given value for a given variable (e.g., high population growth) to particular values of a different variable (e.g., low human capital accumulation).

Importance of the macro to micro alignment

This approach is justified because decision-making under deep uncertainty is often "not based on average or expected values or the most likely outputs, but instead on the consequences of relatively low-probability outcomes." Furthermore, they argue that "there is a well-established trade-off between the internal consistency of scenarios and the risk of being too conservative – for instance, by assuming that the relationship between two variables will remain unchanged over time."

"Top-down" microsimulations are designed to align the micro-level changes with available macro projections. In other words, they impose macro-micro consistency on the microdata. While they can still consider a variety of growth and structural change scenarios depending on the assumed demographic pathway or climate scenario, they attempt to provide a single best-guess estimate based on a single forecast trajectory delivered by the macro models. Reduce the number of possible scenarios to those more likely to occur and ensure theory-based constraints among most (but not all) macroeconomic aggregates, e.g., population aging restricts the economic growth projections, ultimately determining employment and productivity growth in each economic sector. At the other extreme, the *shockwaves* approach, by construction, rejects the idea of "single best forecasts" or "most likely trajectories" and produces thousands of scenarios to construct a wide range of climate impacts and explore the assumptions behind them. They explicitly "disregard macro-economic coherence a priori" (Hallegatte & Rozenberg, 2017) and randomly and independently draw a set of parameters from intervals without considering interdependencies between them. "For instance, nothing prevents us from running a scenario with very high productivity growth in agriculture and no growth in the other sectors of the economy, or a scenario with a skill premium of 1 in services and 5 in industry" (Hallegatte & Rozenberg, 2017). The "Shockwaves" approach explicitly forgo internal consistency between economic fundamentals and micro-simulation to explore many possible futures.

The need to align the micro changes with exogenously determined macro projections is a mixed blessing. On the one hand, it ensures consistency of the micro forecast with the macro counterfactual. On the other hand, it assumes no uncertainties in the available macro forecasts. However, in practice, there is a significant variation in the availability and quality of sensible macro projections among countries. As explained in Section 2, sometimes, we require additional assumptions to link aggregate growth rates to structural change effects in terms of changes in employment and productivity. In extreme cases, there might not be any sensible macro

model available. By randomly drawing a parameter from a range of values, instead of taking macroeconomic forecasts at face value, the "Shockwaves" approach incorporates some of the uncertainties inherent in the long-term macro projection in their simulations.

Plausibility of simulation scenarios

Ironically, they do not require macroeconomic consistency in combining a wide range of random draws of potential drivers, because they draw most potential drivers as if they were independent. The "Shockwaves" do not attribute probabilities or likelihood to our scenarios. Thus, the extensive collection of "scenarios" cannot be used as forecasts or predictions of the future of poverty or as inputs in probabilistic cost-benefit analysis (Hallegatte & Rozenberg, 2017).

However, there is intrinsic plausibility associated with each scenario. Particular shifts in the demographic structure regarding aging and schooling attainment will make certain economic trajectories more likely, determining the likelihood of observing particular employment and wage structures. For example, a favorable draw in human capital accumulation makes scenarios of low fertility population growth more likely (since human capital accumulation is at least associated with lower fertility rates, if not causally related). The importance of aligning microsimulations with macro forecasts and associated demographic trends relates to the plausibility and coherence of seeing those conditions in the future in a given development pathway.

Incorporating household-level climate damages

An explicit limitation for both approaches is related to the possibility of incorporating household-level damages into the microsimulation. The current versions of the "Top-Down" approach applied to assessing climate change impacts on household welfare completely ignore the redistribution effect of climate damages at the household level. In its modeling architecture, climate damages are only accounted for at the macro-level modeling, affecting aggregate growth possibilities, which are subsequently downstream to households through its indirect effect on the distribution of employment and wage opportunities. No micro-level damage function is calibrated for any of the climate hazards. On the other hand, the shockwaves approach is explicitly designed free of macro alignment because it introduces micro-level damages as a critical mechanism determining household income-generating potential and, ultimately, its poverty level. As in the case of the other parameters included in the microsimulation, the inclusion of micro-level damages uses general *ad-hoc* rules that do not necessarily align neither with the knowledge of the geographical distribution of vulnerabilities to climate hazards nor with observed and predicted differences in the distribution of exposures to those hazards. Since the distribution of vulnerabilities is likely to have well-defined geographical patterns and have complex associations with households' levels of exposure to future livelihood resources and other socioeconomic dimensions (such as the household's income diversification strategy), again, given that the "Shockwaves" 's main objective is not to provide an estimate of the future effect of climate change on households' welfare but only reveal the range of worst and best potential outcomes, accounting for the geographical distribution of vulnerabilities to particular climate hazards, its associations with the socioeconomic distribution of exposures and the concurrent, cascading and compounding effects of climate hazards might not consider being significant.

It is crucial to recognize that both "Top-down" microsimulation approaches, the *behavioral* and *reweighting* approaches, emphasize only the growth effect of climate change and climate action. In other words, projected income distribution in the future incorporates only long-term demographic changes (changes in the composition of the population by sex and age distribution, by education level, and by urban residence) and long-term growth and their associated structural change patterns (i.e., changes in the distribution of employment and wages across different types of household). Climate change damages are modeled only at the macro level and affect families only through their effects on growth in the amounts of jobs available in each economic sector and the associated changes in the wage level. Neither of the two models recognizes individual or household climate damages coming from different levels of vulnerability nor differences in exposure to more frequent and more intense climate hazards.

On the other hand, the *Shockwaves* approach recognizes that a pure redistribution effect of climate change is missing because some households will be more affected by some particular risks than others. While the "Shockwaves" approach incorporates some micro-level damages at the individual or household level and simulates a redistribution effect of climate change, the distribution of damages is also simulation is typically *ad-hoc* and independent of both the distribution of vulnerabilities to climate hazards and the distribution of exposure to those hazards will have complex associations with other household dimensions such as the region

where it chooses to live or the diversification of their sources of income.

6 Conclusions

Microsimulation is the only feasible approach to analyzing the household welfare impacts of a wide range of shocks across households within a country. Microsimulations have been very useful for assessing households' welfare fiscal incidence analysis, the effects of liberalization and trade reforms, and, lately, the distributional impacts of the COVID-19 pandemic, among other applications. Given the importance of climate change and the imminent increase in households' vulnerability and exposure to more frequent and intense climate hazards, microsimulation holds the promise of being a valuable tool to analyze the distribution of welfare losses across households not only in the immediate future but in the long run, at least 20 to 30 years from now. The long-term nature of climate change impacts, their associated uncertainty, and, more fundamentally, the heterogeneity across communities and households in vulnerability and exposure to climate damages remain largely unexplored in the microsimulation literature.

This guidance reviews two varieties of *Top-Down* approaches: a conventional *behavioral* approach and a pure *reweighing* approach; and two versions of the *shockwaves* approach. Top-down approaches impose micro-macro consistency to the results of microsimulation in the hope that forecasted changes in a household's welfare are consistent with general equilibrium effects modeled at the macro level. Macro parameters are obtained from external projections such as well-established demographic projections of the distribution of population or economic growth forecasts - and their associated effects on the wage and employment structure simulated using general equilibrium models. Demographic and economic growth pathways and consequences are downstream to households via particular **transmission mechanisms**. In the case of the current state of microsimulations, the labor market. Because in most countries, in particular developing countries, a majority of household income comes from the labor market, microsimulation of climate-change effects relies heavily on growth losses associated with climate damages and their particular impact on the distribution of employment and wage opportunities.

The *shockwaves* approach breaks free from micro-macro alignment to explore not only a single best-guess future associated with particular growth losses in a predefined climate scenario but a wide range of combinations of alternative future pathways and identify the drivers of the more optimistic and more pessimistic outputs in the future. Breaking free from the micro-macro alignment is a double-edged sword. On the positive side, breaking free from strict macro alignment allows microsimulations to explore a broader range of future scenarios regarding demographic and economic pathways and incorporate the effect of *climate-induced damages* due to heat stress or the increasing burden of disease. On the negative side, *ad-hoc* selection of the distribution of demographic or economic parameters will inevitably miss the interactions between future demographic pathways, economic growth trajectories, and their associated structural change effects on the distribution of employment and productivity. Furthermore, since parameters are usually drawn from uniform distributions over a wide range of possible values, we cannot estimate the likelihood of a particular draw of parameters. Very unlike draws (e.g., high education, high population growth, low economic growth, and increases in agriculture employment with lower agricultural wages) are as equally likely as a more plausible combination of draws (e.g., high human capital accumulation, lower population growth, more urbanization, high economic growth, and increases in employment and wages in industry and services). Variations of the shockwaves that use macro parameters as anchors (means) and reduce the range of possible values still use a uniform distribution to draw parameters and, therefore, assume all combinations of values have an equal probability of occurring in the future. If a probability distribution is associated with each outcome. While it might be helpful to depict the full range of possible future, it might be more attractive to pin down a particular best-guess climate impact associated with a small number of climate scenarios, say SSP1 to SSP5. Notice that "Top-Down" microsimulations rely entirely on the availability and credibility of macro projections. If macro models are well calibrated and the assumption behind their trajectories and equilibrium conditions are sound, imposing macro-to-micro general equilibrium alignment makes microsimulations consistent and informative of the simulated future. On the other hand, if there are fundamental flaws in the projections or the macro modeling, the microsimulations will carry on these inconsistencies.

Unfortunately, both models fail to account for the full range of impacts of climate change on household welfare. The "Top-Down" approaches concentrate exclusively on pure growth effects of climate hazards. They

incorporate damage functions and mechanisms only at the macro-level, making all welfare climate effects depend on changes in aggregate growth and their associated changes in employment and wages by economic sector ¹⁹. Making the labor market the only damage mechanism affected by climate damages misses all households' specific vulnerability and exposure to various climate hazards. Furthermore, it also ignores the full range of concurrent, cascading, and compounding effects on their household welfare. At the other extreme, "Shockwaves" microsimulations have attempted to incorporate some damage mechanisms at the micro-level. Still, they have used a *random approach* to assign household damages. A random approach to micro-level climate damages may be problematic if there are clear patterns that relate to vulnerability to particular regions or communities and that relate exposure to such hazards to certain socioeconomic characteristics. Specific areas and communities will suffer from more frequent and intense hazards, e.g., sea level rise will more likely affect communities on the coast, and specific households will have a higher share of their livelihood assets exposed to such hazards, e.g., agricultural workers are more likely to work outdoors in high demanding context so that they are more likely to be exposed to heat stress effects on their productivity of labor. Finally, not using simulated socioeconomic information to assign vulnerable and exposed to climate hazards will miss the significant correlation between different threats and their concurrent, cascading, and compounding effects on the welfare of particular households.

Comparing both "Top-Down" approaches, the *behavioral* and the pure *reweighting* approaches, it is difficult to ascertain the superiority of one over another. Validation of both approaches in hindcasting exercises under different country contexts regarding development data availability and quality still needs to be determined. There might be some gain in using behavioral models when the specifications are done in a way that guarantees a more accurate approximation to the changes we want to approximate. However, preliminary out-of-sample evaluation of the accuracy of their predictions in terms of employment status and economic sector show that more than commonly available variables are needed to capture behavioral relationships better than the random allocation for most occupational choices. Furthermore, similar out-of-sample evaluation exercises to assigning a counterfactual income show further complications because traditional linear approaches can not account for more than 1/3 of the total variation, and we are interested in forecasting a distribution. The problem is further complicated when annual labor earnings are not the objective variable of the surveys, and the variable has to be approximated by declared monthly earnings or, worst when imputation or assignment rules have to be used to assign to each working member an "imputed labor income."

On the other hand, the practical implementation of reweighting methods is subject to caveats. The most appropriate number of population partitions will depend on the sample size and original population distribution among the different segments and the chosen demographic pathways. Conservative but unlikely assumptions about the trajectories in education, such as the no change in enrollment or the constant enrollment rates, may be easier to calibrate but are not necessarily the more realistic. Constant historical trend assumptions Constant Education Trend (CET) assumption allows for continuous improvements in educational attainment and is more difficult to calibrate but may be more likely and appropriate.

(ii) specific areas of divergence across methods and reasons for deviation,

The balance between available time, expected quality, and budget considerations.

6.1 Recommendations for policy applications

From our review of approaches for the assessment of households' welfare impacts of climate change and climate policy, several recommendations for applied works come to our mind,

1. **Choosing and cleaning baseline data.** As () emphasize, selecting and pre processing the most appropriate baseline for the objective is a fundamental first step toward informative microsimulations. Microsimulations main ingredient is good quality data on labor earnings. In developing countries, particularly in contexts with high levels of self-employment and informality, labor earnings information is prone to non-response, out-of-range values, and different measurement errors. Furthermore, in some contexts, declared labor earnings are captured concerning the last month. Using usual, expected, or filtered measures of annual labor earnings that clean obvious measurement errors and smooth out noisy survey aggregates can go a long way. Furthermore, in a context where labor earnings are not available -at least at the individual level, using procedures such as the Residual consumption-based income

¹⁹Demographic pathways are usually kept constant in the microsimulations.

imputation that assigns consumption to employed individuals based on their productive characteristics and segmentation patterns of the labor markets can also improve the quality of the simulations and make microsimulation numbers consistent with official poverty estimates available for the countries. Documenting these procedures not only enhances the transparency and replicability of the exercises but may also help interpret some of the results.

2. **Examine the consistency of employment and labor income data.**
3. **Choosing demographic pathways.** Alignment of micro to macro baseline data requires consistency of assumptions between macro and micro projections.
4. **Choosing between static and dynamic aging strategies.**
5. **Explore sensitivity of the simulation results to alternative scenarios.**

6.2 Suggested areas of future research

1. **Documenting and validating procedures to assign individual labor earnings from consumption or smooth labor earnings.** Performance of different methods across different country characteristics and
2. **Validation of "Top-Down" approaches.** performance of different methods across different country characteristics and
3. **Incorporate redistribution impact into the "Top-Down" approach.** This implies investing in data availability and processing of climate drivers' calibration of macro and micro-level damage functions begin harmonizing and analyzing data on households' exposure to climate hazards. (Hernani-Limarino, 2023) proposes a simple procedure for downstream macro-level impacts to micro-level effects based on the construction of household-level damage functions, consistent with the macro-level damage function used in *Top-Down* micro simulations. It also illustrates how this adjustment can be used to calculate welfare loss functions for each climate hazard and recalculate the welfare impact of climate change as the sum of two components: a pure growth effect and a redistribution effect.
4. **Perform an inventory of survey questions related to risk analysis.** Given the importance of climate risk analysis, it might be essential to inventory available questions, design the most appropriate exposure questions, and use new household data collections to test and improve them.

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A Application of the reweighting approach in the Sahel region

The Country Climate and Development Report (CCDR) for the G5 Sahel region [World Bank (2023)] illustrates how the reweighting approach was used to analyze the distributional effects of climate change and their associated impacts on poverty and inequality. It is important to highlight that long-term microsimulations are the final step of a process that involves:

1. Defining a set of development trajectories
2. Defining a set of climate change scenarios
3. Defining a collection of climate hazards to analyze
4. Producing (average) damage pathways
5. Introduce damage pathways into a CC-MFMod to forecast their effect on macroeconomic aggregates, in particular economic growth.
6. Producing baseline and counterfactual distributions and their associated poverty and inequality impacts, using long-term microsimulation.

A.1 Development trajectories

The first step in estimating the welfare impacts of climate change using a top-down approach to long-term microsimulations is to define a development trajectory. In other words, define a baseline trajectory that includes pathways of demographic changes, economic growth, and structural transformation. In the case of the CCDR for the Sahel G5, demographic changes were defined using Medium Variant UN population projections (by age and sex cohort), education projections under a constant enrollment rate assumption (that froze education progression at the most recently observed school enrollment level), and three growth scenarios with different assumptions on the average annual growth in GDP per capita that will occur over the next 30 years (2021-2050) and the speed of structural transformation: (i) a "lower growth/stagnation" scenario that assumes stagnation of per capita income with no structural transformation; (ii) a "medium growth" scenario that assumes a growth performance in each country that matches the growth rate observed during the last 30 years (1991-2020); and (iii) a "higher growth" scenario that assumes a growth rate that doubles the medium growth scenario and resembles the economic take-off experience of former low-income countries such as Bangladesh or Vietnam.

A.2 Define Climate Change Scenarios

To account for alternative climate futures in the Sahel, six climate scenarios were defined for each of the five countries²⁰. The first three scenarios accounted for variations in emissions, while the last three accounted for variations across climate models. Table 1 describes the rationale behind the climate scenarios.

A.3 Define Climate Hazards to include and their damage mechanisms

Parallel to this, a set of climate channels or hazards were identified for each climate scenario. The relevant climate hazards are described in Table 2.

A.4 Modeling macro damage functions

Once scenarios and hazards were defined, macro-level damage functions were constructed for each climate hazard, combining geographically disaggregated global climate data with biophysical models. Damage models provide the average GDP economic impact of each climate hazard for each damage and climate scenario²¹.

²⁰Climate scenarios were defined exploring data from the World Bank's Climate Change Knowledge Portal (CCKP) for 29 General Circulation Models (GCMs) from the Coupled Model Intercomparison Project 6 (CMIP6) suite of IPCC model outputs

²¹Damage modeling usually requires additional data to calculate the biophysical effects and produce projections of average annual shocks from 2021 to 2050.

A.5 Modeling growth effects of climate damages using a CC-MFMod

To assess the impact of climate change on overall GDP and macroeconomic aggregates from 2021 to 2050, annual average damages from each of the impact channels are incorporated as "shocks" into the modified CC-MFMod. The Climate Change Macro-Fiscal Model (CC-MFMod) is an advanced iteration of the standard MFMod, typically employed for core macro-modeling across our focus countries. As in most Top-Down approaches, macro modeling is the main instrument for analyzing the impact of climate change hazards, at the individual and combined level, on economic growth and other macroeconomic aggregates.

Overall, the impact of these channels on economic output varies significantly by climate scenario and country. Figure X1 presents the projected economic losses relative to baseline GDP up to 2050. The data reveals a trend of escalating output losses across these countries due to climate change, with the severity of the impact increasing over time. A key observation is that these losses are markedly more severe under the dry/climate pessimistic scenario than the wet/climate optimistic scenario. In the optimistic scenario, rainfed crop and livestock yields demonstrate beneficial impacts, which help offset the adverse effects observed in other areas.

By 2050, under the wet-climate optimistic scenario, countries like Niger and Mali are expected to see their annual GDP decrease by 2.2% to 6.4%. Under the dry-climate pessimistic scenario, this decrease is projected to be more drastic, ranging from 6.8% to 11.9% in countries such as Burkina Faso and Niger. These reductions are substantial and can erase annual real GDP per capita growth.

When analyzing GDP losses about the baseline, the low growth scenario consistently shows higher losses, especially under the dry/climate pessimistic scenario. A trend that is particularly evident in countries like Chad, Mali, and Niger. In these low-growth scenarios, where minimal structural transformation occurs and the economy heavily relies on agriculture, the negative impacts of climate shocks are amplified. In contrast, medium and high-growth scenarios mitigate these impacts, reflecting a shift from agriculture.

Despite the anticipated damages from climate-related shocks, projections indicate that annual GDP and GDP per capita will be significantly higher after accounting for climate change impacts in the medium and high growth scenarios. This is because these scenarios' baseline levels (excluding climate shocks) are much higher than those in the lower growth scenarios. Figure X further illustrates this point by showing that for all countries, the projected 2050 GDP per capita in medium and high growth scenarios—even factoring in climate change shocks under the dry/pessimistic climate scenario will still exceed the figures in the low growth scenario baseline without climate shocks. This highlights the vital role of robust underlying economic growth in offsetting climate change's adverse effects and fostering an increase in per capita income.

A.6 Analyze poverty and distributional effects of climate change using microsimulation

To analyze poverty and distributional impacts under different macroeconomic and climate scenarios, the Sahel G5 CDDR combines the latest living standards or household data with the above macroeconomic growth forecasts using a reweighting approach to align microsimulation with macro forecasts. The resulting counterfactual distributions were used to generate poverty and inequality trends for each country, each growth trajectory, and each climate scenario. Finally, climate impacts were obtained by comparing the counterfactual trajectories with their corresponding baselines ²².

Climate change impacts on poverty

Figure X illustrates how the combined effect of hazards on economic growth translates into an increase in poverty measured as deviation from the baseline for all five countries and all the growth and climate scenarios. Poverty will increase throughout 2050 under the dry/pessimistic climate scenarios. By 2050, under a medium growth and dry climate scenario, the poverty rate will increase relative to the baseline by 0.2 percentage points in Mauritania, 4.7 percentage points in Mali, 5.3 percentage points in Burkina Faso, 8.6 percentage points in Niger, and 10.6 percentage points in Chad. For the region as a whole, it is estimated that by 2050, there will be an increase in the poverty rate from 27 percent projected under the medium growth baseline to 29 percent under the wet/climate optimistic scenario and 34 percent under the dry/climate pessimistic scenario. Like the poverty rate, the number of additional poor relative to the baseline increases in all countries

²²Baseline development trajectories (low, medium, and high) for each economy extending up to 2050 assume no additional climate change impacts beyond those already experienced up to 2020. They serve as a reference point for comparing outcomes from climate change-related shocks and policies. Correspondingly, the three counterfactual trajectories incorporating climate shocks are developed for each growth scenario up to 2050.

under all scenarios. The number of additional poor is also higher in 2050 and much worse for the dry climate than the wet climate scenario. By 2050, under a medium growth and dry climate scenario, the number of poor will increase relative to the baseline by 0.02 million in Mauritania, 2.21 million in Mali, 2.7 million in Burkina Faso, 3.34 million in Chad, and 5.23 million in Niger.

The climate change impacts on inequality

With or without climate change, growth is not expected to be pro-poor, and the Gini increases over time. The impact of climate change on the Gini is marginal in Mauritania and Mali. However, in Burkina Faso, Niger, and Chad, the dry climate scenario leads to a more significant increase in inequality. In contrast, the wet climate scenario leads to a more minor increase in inequality relative to the baseline. It is essential to notice that this increase is not caused by a higher exposure of poor people to shocks (natural disasters, security, health, etc.) or less participation from government redistributive efforts because neither of these two factors is included in the modeling. Furthermore, per capita consumption is expected to decline for the poorest decile compared to the wealthiest decile for all the countries under the dry climate scenario by 2050. The poorest are affected more under the dry climate/pessimistic climate scenario as the heat stress-labor productivity impact channel affects more outdoor workers (generally in agriculture and poorer). Other channels, such as rainfed crop and livestock yields, mainly affected by dry climate scenarios, are in agriculture (where there is a higher share of poor households). In addition, in the face of drought, the poor have less access to irrigation infrastructure and tend to rely on detrimental coping mechanisms such as selling livestock and other durable assets.

A.7 Data and knowledge limitations

It is important to recognize that the simulation analysis results do not provide definitive answers. Results are inherently conditional on a series of assumptions. Therefore, caution is advised when using specific numbers to assess the challenges and trade-offs involved in climate change. Some key caveats in the Sahel G5 modeling approach include:

- Partial set of climate hazards. The modeling approach includes only specific impact channels, meaning that the GDP impacts reported are partial and not comprehensive of all potential risks. Our model has yet to include significant potential GDP losses from climate-induced migration and conflict.
- Assumptions of no-adaptation. The welfare impacts are modeled on the assumption of no significant adaptive response. For example, the model assumes a constant crop mix in each country without considering potential adaptation policies taken by households.
- Omission of climate change non-linear effects. The current damage modeling approach does not account for climate hazard's compounding and cascading nature. Additionally, they overlook the heterogeneous spatial distribution of these hazards and the inequality in exposure distribution. Omitting these factors could significantly underestimate climate change's impact on poverty and inequality.
- Exclusion of mitigation effects in the Sahel Region. Although the Sahel region contributes minimally to global greenhouse gas emissions, it is essential to point out that the analysis does not account for adverse welfare effects induced by labor reallocation due to the transition toward more sustainable energy sources in this region.
- Uncertainty in development paths and scenarios. While the modeling explores various development paths and climate scenarios, each path and climate scenario comes with significant uncertainties. For example, in the case of development scenarios, they include but are not limited to the evolution of fertility, education, structural effects of growth, and climate outcomes, all of which add complexity to the predictions.

B Linkage variables for "Top-Down" approaches

Mechanic simulations project Behavioral simulations attempt to predict what would be the (optimal) response of a household defining some behavioral rules (such as rational utility maximization given a set of preferences between consumption and leisure)

B.1 Linking variables I: Demographic targets

Alignment and target distributions

B.1.1 Population projections

See, for instance, the United Nations' regular publication, United Nations Population Prospects (<https://population.un.org/wpp/>)

B.1.2 Schooling distribution

For instance, long-term projections of schooling levels of the future population for most countries in the world are produced by the International Institute for Applied Systems Analysis (see <https://iiasa.ac.at/models-tools-data/wcde>) as per methods developed by [(Lutz et al., 2018)].

Another key LAV parameter usually overlooked in forecasting scenarios is the trend in human capital accumulation and the future distribution of education and skills. Trends in human capital accumulation are critical determinants for every other key LAV parameter in the "Top-Down" and the Shockwaves approaches. Again, neither the GIDD incorporate the uncertainty about the evolution of human capital accumulation and its consequences on welfare, nor does the *Shockwaves* distinguish the dependency of alternative human capital accumulation paths on the development of other vital parameters such as changes in population growth and composition, dependency rates, labor market participation, employment reallocation, relative wages or total economic growth. To project human capital accumulation, the GIDD assumes a Constant Enrollment Rate (CER), i.e., the most recently observed educational enrollment rates are frozen, and no further improvement is allowed. The CER assumption is pessimistic because it implies that, in the long run, the education distribution will stagnate. All changes in the distribution of education occur because younger cohorts are better educated (and usually more numerous). A more optimistic view of the human capital accumulation process allows schooling attainment to increase to match the observed trend in human capital accumulation by cohort - with marginal increases allowed to diminish over time, this more optimistic Constant Education Trend (CET) assumption provides for continuous improvements in educational attainment (See Figure [EduConstantTrend] for an illustration.). An even more optimistic forecast can be obtained if we assume that countries' human capital accumulation will follow South Korea's rapid expansion - a.k.a. the Fast Track (FT) scenario. Different human capital accumulation trajectories have significant effects on all other vital LAVs. Figure [PopProj] illustrates the differences in population projections for the African continent under the CER, CET, and FT scenarios. Figure [PCIProj] shows the ceteris paribus effect of the CER and CET scenarios on the forecast of the per-capita income distribution for Cameroon.

Education categories definition

1. No education (E1) No formal education or less than one year of primary
2. Primary (E2) Uncompleted primary completed primary (ISCED 1) and uncompleted lower secondary
3. Secondary (E3) Completed lower secondary (ISCED 2), uncompleted and completed higher secondary (ISCED 3/4), and uncompleted tertiary education
4. Tertiary (E4) Completed tertiary education (ISCED 5/6)

Assumptions

1. The fast track (FT) scenario is extremely ambitious; it assumes that all countries expand their school systems at the fastest possible rate, comparable with best performers in the past, such as Singapore and South Korea.
2. The global education trend (GET) scenario is moderately optimistic. It assumes that countries will follow the average path of school expansion that other countries already somewhat further advanced in this process have experienced.

3. The constant enrollment rate (CER) scenario assumes that countries only keep the proportions of cohorts attending school constant at current levels.
4. The most pessimistic scenario, constant enrollment numbers (CEN), assumes that no more schools are being built and that the absolute number of students is kept constant, which under conditions of population growth means declining enrollment rates.

B.1.3 Urban/Rural distribution